Problems

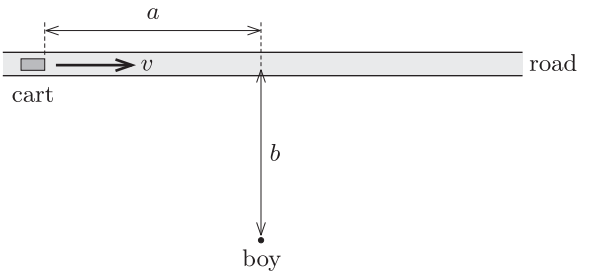
P1 The trajectories of two bodies moving with non-relativistic constant speeds are parallel in a particular inertial reference frame.

a) Is it possible to choose another inertial frame of reference in which the two trajectories cross each other?

b) If such a frame can be found, and the bodies are started with suitable initial conditions, then it could be arranged that they reach the crossing point at the same time. How can this be consistent with the parallel trajectories observed in the first frame of reference?

P2 Ann is sitting on the edge of a carousel that has a radius of 6 m and is rotating steadily. Bob is standing still on the ground at a point that is 12 m from the centre of the carousel. At a particular instant, Bob observes Ann moving directly towards him with a speed of 1 m s^{-1}. With what speed does Ann observe Bob to be moving at that same moment?

P3 A cart is moving on a straight road with constant velocity $\boldsymbol{v}$. A boy, standing in an adjoining meadow, spots the cart and hopes to get a ride on it. In which direction should he run to catch the cart? Solve the problem generally: denote the speed of the cart by v, the maximal speed of the boy by u, and take the initial positions of the cart and boy to be as shown in the figure.



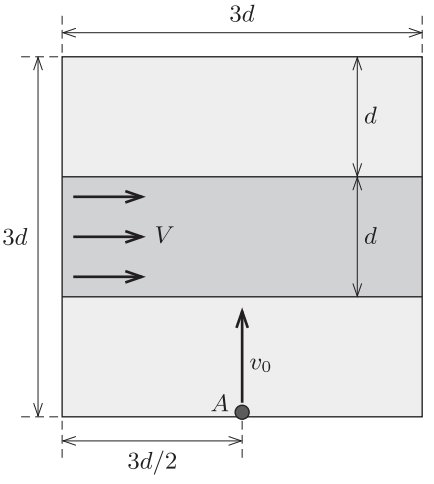
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P4* A group of Alaskan gold prospectors reach a wide straight river that flows with uniform speed v. What immediately catches all their eyes is a huge gold nugget lying on the further bank, directly across the river. The laws governing prospecting in Alaska state that the first person to reach any particular place has the right to establish a mine there; speed is of the essence!

Joe, one of the prospectors, has a canoe, which he can paddle in still water at the same speed u as he can hike along a river bank. What course of action should he take if u/v is (a) smaller than or (b) larger than a certain critical value? Assume that Joe first paddles across the river (in a straight line) and then, if necessary, hikes along the bank to reach the nugget.

P5 The top surface of a horizontal laboratory table is a square of side $3d = 3$ m. Running centrally across the table, and parallel to one of its sides, is a conveyor belt consisting of an endless rubber band of width $d = 1$ m, which moves with a constant velocity $V = 3$ m s^{-1}. The height of the belt's upper surface exactly matches that of the static part of the table.

A small, flat disc is placed at the middle of one of the edges of the table (at the point A shown in the figure), and the disc is hit so that it starts sliding with velocity $v_0 = 4$ m s^{-1} at right angles to the belt. The friction between the disc and the static part of the table is negligible, while the coefficient of kinetic friction between the disc and the rubber band is $\mu = 0.5$.



Where does the disc leave the table?

P6* A boy is running north, with a speed of $v = 5$ m s^{-1}, on the smooth ice cover of a large frozen lake. The coefficient of friction (both kinetic and static) between the tread of his trainers and the ice is $\mu = 0.1$. For the sake of simplicity, assume that the normal force he exerts on the ice, which in reality changes with time, can be substituted by its average value.

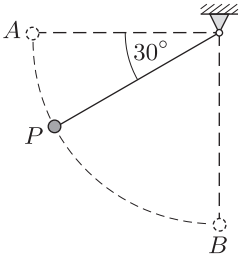
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a) What is the minimal time that he needs to change direction, so that he is running east with the same speed v?

b) Find the boy's trajectory during the turn in this optimal case.

P7* A simple pendulum is released from rest with its string horizontal. What kind of curve is the locus of the end of its acceleration vector?

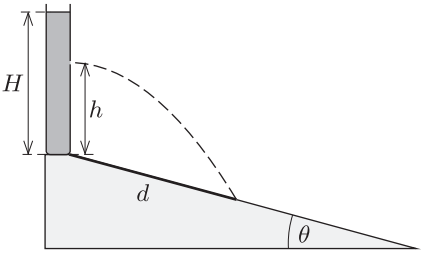
P8 A simple pendulum is released from rest with its string horizontal. Which of the two arcs, *AP* and *PB* as defined in the figure, will its bob cover in a shorter time?



P9 The trajectory of a projectile with initial speed v_0 is parabolic in a vacuum (e.g. on the Moon). How far is the focus of this parabola from the launch point? What initial angle of elevation of the projectile is needed if the focus is to be at the same altitude as the launch point?

P10* Point-like objects are thrown with an initial speed of v_0 in various directions from the top of a tower of height h. If the air resistance is negligible, what is the maximum distance from the foot of the tower that they can reach?

P11* At the top of a long incline that makes an angle θ with the horizontal, there is a cylindrical vessel containing water to a depth H. A hole is to be drilled in the wall of the cylinder, so as to produce a water jet that lands a distance d down the incline. How far, h, from the bottom of the vessel should the hole be drilled in order to make d as large as possible? What is this maximum value of d?



P12 Because of the finite exposure needed, in a side-on photograph of the front wheel of a moving bicycle, the spokes seem blurred. However, there will be some

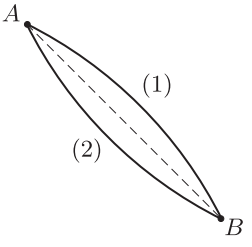
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apparently sharp points in the picture. Where are these sharp points? For the sake of simplicity, suppose that the bicycle spokes are radial.

P13* Investigate the form of the image of a spoked bicycle wheel as recorded by a photo-finish camera. Such cameras use very narrow strip photography, electronically capturing a vertical cross-section of the sequence of events only on the finish line; every part of each body is shown as it appeared at the moment it crossed the finish line. The horizontal axis of the image represents time; anything stationary on the finish line appears as a horizontal streak. In a conventional photograph, the image shows a variety of locations at a fixed moment in time; strip photography swaps the time and space dimensions, showing a fixed location at a variety of times. For the sake of simplicity, suppose that the spokes of the bicycle are radial.

P14** A cartwheel of radius 50 m has 12 spokes, assumed to be of negligible width. It rolls along level ground without slipping, and the speed of its axle is 15 m s^{-1}. Use a graphical approach to estimate the minimal speed a crossbow bolt, 20 cm long, must have if it is to pass unimpeded between the spokes of the wheel? Neglect any vertical displacement of the bolt.

P15 A small bob can slide downwards from point A to point B along either of the two different curved surfaces shown in cross-section in the figure. These possible trajectories are circular arcs in a vertical plane, and they lie symmetrically about the straight line joining A to B. During either motion the bob does not leave the curve.



a) If friction is negligible, along which trajectory does the bob reach point B more quickly? How do the final speeds for the two paths compare?

b) What can be said about the final speeds if, although friction is not negligible, the coefficient of friction is the same on both paths?

P16 On a windless day, a cyclist 'going flat out' can ride uphill at a speed of $v_1 = 12$ km h^{-1}, and downhill at $v_2 = 36$ km h^{-1}, on the same inclined road. What is the cyclist's top speed on a flat road if his or her maximal effort is independent of the speed at which the bike is travelling?

The rather imprecise term 'effort' could be interpreted scientifically to mean either

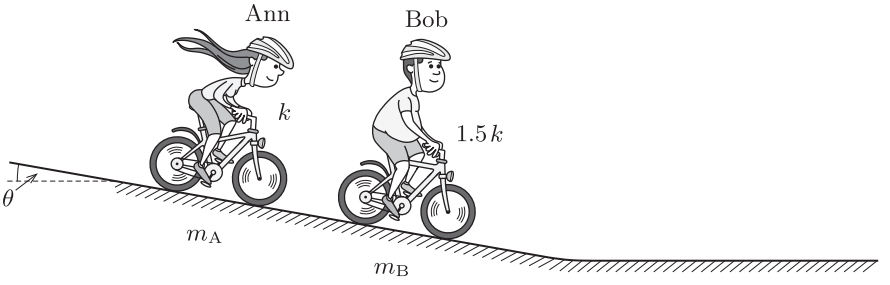
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a) the magnitude of the *force* exerted on the pedals by the rider (which is then transmitted to the wheels via the crank arms, sprockets and chain), or

b) the rider's mechanical *power*.

Solve the problem for both interpretations.

P17* Ann and Bob arrange a 'free-wheel bike race' on a very long slope that makes an angle θ with respect to the horizontal. They have the same type of bicycle, and neither of them pedals during the race. The total mass of Ann and her bike is $m_A = 60$ kg, whereas the corresponding figure for Bob is $m_B = 110$ kg. Because he is overweight (and 'out of shape'), the air drag on Bob is one-and-a-half times larger than that on Ann when they have equal speeds. Which one is going to coast further on the horizontal road at the bottom of the slope?



Assume that their decelerations are due to air drag (proportional to the square of the speed), friction at the bearings hub and rolling friction. The latter two effects should be treated as a sort of kinetic friction, with an 'effective frictional coefficient' of μ.

P18 A small feather with vanishingly small mass is attached to one end of a riding crop by a flimsy thread. The crop is then rotated steadily about an axis passing through its other end and perpendicular to it. What is the trajectory of the feather?

P19* A small pearl moving in deep water experiences a viscous retarding force that is proportional to its speed (Stokes' law). If a pearl is released from rest under the water, then it soon reaches its terminal velocity v_1, and continues sinking with this velocity.

In an experiment, such a pearl is released horizontally with an initial speed v_2.

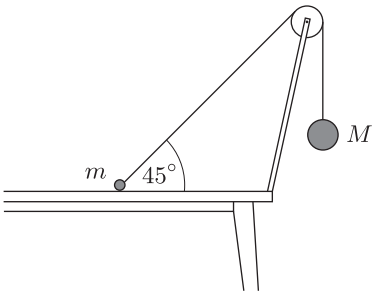
a) What is the minimal speed of the pearl during the subsequent motion?

b) In which direction should the pearl be projected, with the same initial speed v_2 ($< v_1$), in order that its speed increases monotonically during its descent?

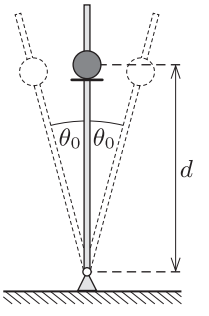
P20 Two spherical bodies, with masses m and M, are joined together by a light thread that passes over a table-mounted pulley of negligible mass. Initially they are held in the positions shown in the figure; then, at a given moment, both of them

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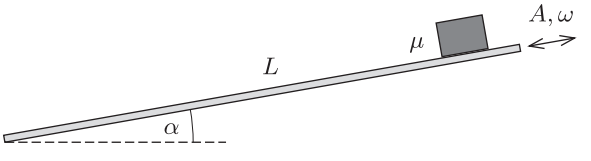
are released. Mass M is many times – say, one thousand times – larger than mass m. The friction between the smaller ball and the surface of the table is negligible. Will the lighter ball be lifted from the table-top immediately after the release?



P21* A small smooth pearl is threaded onto a rigid, smooth, vertical rod, which is pivoted at its base. Initially, the pearl rests on a small circular disc that is concentric with the rod, and attached to it at a distance d from the rotational axis. The rod starts executing simple harmonic motion around its original position with small angular amplitude θ_0 (*see* figure). What frequency of oscillation is required for the pearl to leave the rod?

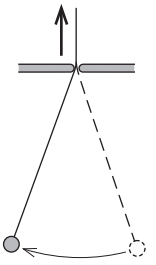


P22** The plane of a flat, rigid board of length $L = 6$ m makes an angle of $\alpha = 10^\circ$ with the horizontal, and a small rectangular block is situated at the top of this incline. The board starts vibrating with simple harmonic motion in the direction parallel to a line of steepest descent; the amplitude of the motion is $A = 1$ mm, and its angular frequency is $\omega = 500$ rad s^{-1}. The coefficients of kinetic and static friction between the block and the board are both $\mu = 0.4$. Estimate how long the block, which does not topple over during its motion, takes to reach the bottom of the incline.

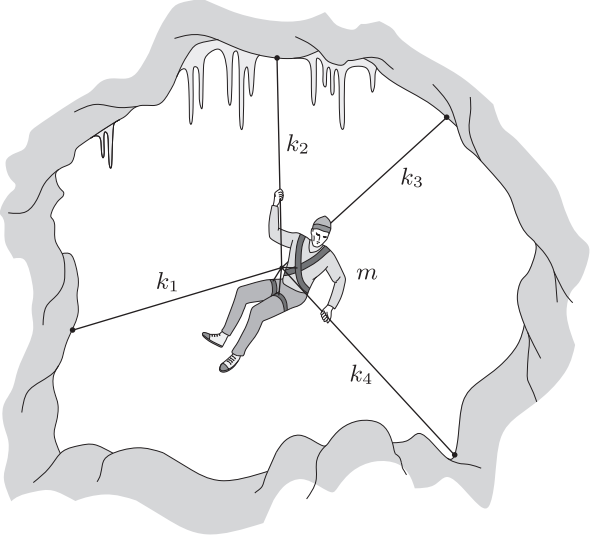


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P23** The cord of a swinging simple pendulum passes through a small hole in a ceiling and into a loft above. There, a scientist's assistant holds the loose end of the cord and pulls it up very slowly (*see* figure). Does the linear amplitude (the largest horizontal excursion) of the pendulum change? If so, how?



P24* A mountaineer (a former circus artist) has to spend the night on the (vertical) side of a high mountain. So, as shown in the figure, he clamps himself to four carabiners fixed to the rock face using four extraordinarily flexible springs. The masses of the springs and their unstretched lengths are negligible, and their spring constants are $k_1 = 150\ \mathrm{N\ m^{-1}}$, $k_2 = 250\ \mathrm{N\ m^{-1}}$, $k_3 = 300\ \mathrm{N\ m^{-1}}$ and $k_4 = 400\ \mathrm{N\ m^{-1}}$. The mountaineer can be considered – for the sake of simplicity – as a point-like body with a mass $m = 70$ kg.

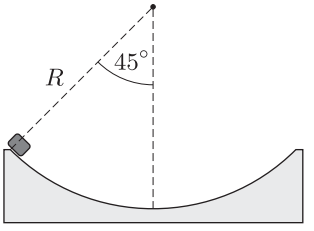


What is the mountaineer's period of oscillation if he is displaced from his equilibrium position and then released?

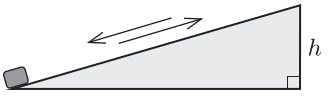
P25 A small rubber eraser is placed at one edge of a quarter-circle-shaped track of radius R that lies in a vertical plane and has its axis of symmetry vertical (*see* figure); it is then released. The coefficient of friction between the eraser and

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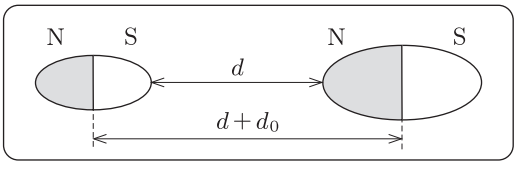
the surface of the track is $\mu = 0.6$. Will the eraser reach the lowest point of the track?



P26 A box of mass 1 kg is placed on an incline on which it does not spontaneously start to slide. It is pulled up, and then down, the incline very slowly in such a way that the traction force is always parallel to the slope (*see* figure). The total work done is 10 J. What is the maximum height h of the incline? Assume that the coefficients of static and kinetic friction are equal.



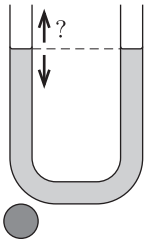
P27* Two permanent magnets are aligned on a horizontal, very slippery tabletop with a gap of length d between them; because of their finite sizes, their centres of mass are $d + d_0$ apart (*see* figure). The magnets are held in such a position that the net force between them is attractive, and there are no torques generated.



If one of the magnets is held firmly in position and the other is released, then the two collide after 0.6 s. If the roles are reversed, then the time interval between the release and the collision is 0.8 s. How long would it take the two magnets to collide, if both were released simultaneously?

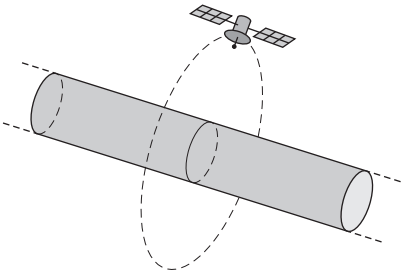
P28* A U-shaped tube contains liquid that initially is in equilibrium. If a heavy ball is placed below the left arm of the tube, how do the liquid levels in the two arms change?

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P29** We wish to produce the maximum possible gravitational acceleration at a given point in space, using a piece of plasticine[1] of uniform density and given volume. Into what shape should the plasticine be moulded?

P30 One of the planets of a star called 'Noname' has the shape of a long cylinder. The average density of the planet is the same as that of the Earth, its radius R is equal to the radius of the Earth, and the period of its rotation around its long axis is just one day.



a) The first cosmic velocity $v_{c,1}$ is the speed of a satellite in a stable orbit just above the planet's surface. How large is it for this planet?

b) What is the altitude of a 'geostationary' communications satellite above the surface of this 'sausage' planet?

c) What can be said about the second cosmic velocity (the escape velocity) for this planet?

P31 The Examining Institute for Cosmic Accidents (EXINCA) sent the following short report to one of its experts:

> One of the exploration space ships belonging to the titanium-devouring little green people has found a perfectly spherical, homogeneous asteroid; it has no atmosphere, but is made of pure titanium. As part of the preparations for mining, a straight tunnel was constructed, and railway lines were laid in it. The length of the tunnel was equal to the radius of the asteroid, with both ends on the latter's surface. Unfortunately, although its braking system was on and locked, one of the mine wagons slipped into the shaft at one end of the tunnel. Initially it speeded up, but later it gradually slowed down, reversed and finally stopped exactly in

[1] Although this word has passed into common usage, technically it is a registered trademark for Plasticine.

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the middle of the tunnel. Just before it reversed, the wagon came very close to running down the mine captain, who was standing on the track.

EXINCA asked the expert (you) to obtain numerical values for the following:

a) how far along the tunnel the mine captain was standing,

b) the coefficient of kinetic friction between the wheels of the mine wagon and the rails,

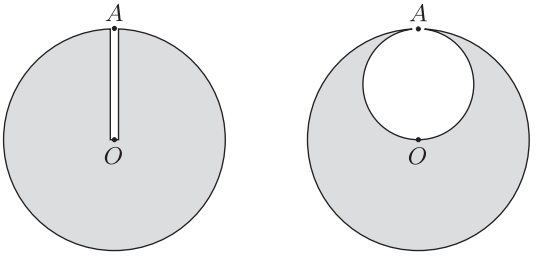
c) the total time of the wagon's motion.

Assume that the volume of the tunnel is negligible compared to that of the asteroid.

P32 A space ship carrying titanium-devouring little green men has found and landed on a perfectly spherical planet of radius R. A narrow trial shaft has been bored from point A on the surface of the planet to O, its centre; this has confirmed that the whole planet is made of homogeneous (edible) titanium. In addition, according to the measurements made, the temperature inside the narrow shaft is constant, and equal to T_0. The planet has an atmosphere with molar mass M, and the atmospheric pressure at its surface is p_A.

a) Find the air pressure at the bottom of the shaft.

After the exploratory drilling, work has continued, and the little green men have started secret excavation of the titanium, as a result of which they have formed a spherical cavity of diameter AO inside the planet, as illustrated in the figure. The excavated titanium is being transported away using expendable cargo space craft. Air from the atmosphere has moved to fill the cavity and, as a consequence, the pressure at the access point A has decreased from p_A to p'_A.



b) Assuming that the temperature everywhere inside the cavity is the same as it was in the shaft, how has the atmospheric pressure at O changed?

P33 Mr Tompkins[2] visited Wonderland in his dream, where the laws of physics are almost the same as we know them – except that gravity deviates 'slightly' from Newton's well-known law. Awakening with a start, he remembered

[2] He is the eponymous main character in the physicist George Gamow's book, *Mr Tompkins in Wonderland*, first published in 1940.

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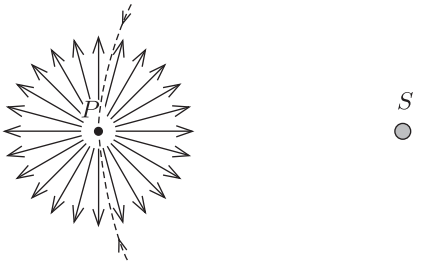
that several planets orbit the only 'sun' in Wonderland, and that these planets obey the following three 'Kepler's laws':

1. Planets travel in elliptical orbits with the sun at the *centre* of the ellipse.
2. A line drawn from a planet to the sun *sweeps* ... (unfortunately, Mr Tompkins forgot how to continue).
3. The orbital periods of all the planets (independent of the sizes of their major and minor axes) are the *same*, namely one 'year'.

How does the gravitational law look in the physics textbooks of Wonderland, and what is the missing part of the statement of 'Kepler's second law'?

P34 In the absence of an atmosphere on Earth, what would be the maximal and minimal impact speeds with which a comet, which is orbiting the Sun, can strike the Earth?

P35** Two comets with identical masses and speeds are found by astronomers to be approaching the Sun along parabolic trajectories that lie in the same plane. The comets collide at their common perihelion P (the point in their trajectories that is nearest to the Sun S), and break into many pieces that then go in all directions, but with identical initial speeds (*see* figure).



What shape is the envelope of the subsequent trajectories of the pieces of debris?

P36** We aim to make a space probe launched from Earth leave the Solar System with the help of a single gravitational slingshot, which utilises the relative movement and gravity of one of the planets that orbit the Sun. In astronomical units (the mean Sun–Earth distance = 1 AU), how far from the Sun would the 'ideal' planet be if the initial launch speed of the probe is to be kept to a minimum?

When solving the problem, make the following approximations:

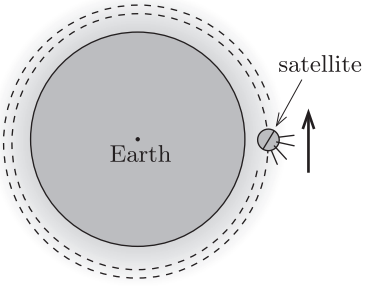
- The orbits of the planets are circles all lying in the same plane.
- Near a planet, it is sufficient to take into account only the gravity due to that planet.
- Far from all planets, only the Sun's gravity is relevant.

Does a real planet exist at or near the optimal orbit?

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P37* Because of the effects of air drag, abandoned satellites, at the end of their useful lives, lose energy in the upper layers of the atmosphere, before finally burning up when they reach the denser lower layers. It can be shown that satellites originally moving along circular trajectories will continue to travel in approximately circular orbits, with their orbital radii slowly decreasing.

A half-tonne satellite is orbiting the Earth on a roughly circular orbit when it is abandoned. The drag acting on this particular satellite can be expressed as $c\varrho v^2$, where $c = 0.23\ \mathrm{m}^2$, ϱ is the density of air at the altitude of the satellite and v is the speed of the satellite.



a) Does the satellite brake or accelerate as a result of the air drag? How can your answer be explained from the point of view of dynamics?

b) A simple connection can be found between the drag force and the tangential acceleration of the satellite. What is it?

c) What is the density of air at an altitude of 200 km, if in this region the orbital radius of the satellite decreases by 100 m during a single revolution?

P38** It is a well-known fact that the Moon always shows, more or less, the same face to the Earth. This curiosity is not a coincidence, but a straightforward consequence of the tidal forces acting between the Earth and the Moon. Over time, tidal forces continuously slowed the Moon's rotation about its own axis until the period of that rotation became equal to the Moon's orbiting period around the Earth. It is for the same reason that the Earth's rotation around its own axis is continually slowing, and the orbital speed of the Moon is further decreasing.

a) Estimate the ratio of the rates of decrease of the Earth's and Moon's kinetic energies.

b) During the Apollo Program (flights 11, 14 and 15), retro-reflectors (arrays of corner-cube laser mirrors) were placed on the Moon. According to the extremely accurate laser ranging measurements thereby made possible, the

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Moon's linear distance from Earth is currently increasing at a rate of 3.8 cm per year. Using this measured datum, estimate the change in the length of an Earth day during a year.

c) If the Earth–Moon system continued its motion undisturbed, then, as a result of the braking effect of the tidal forces, after a sufficiently long time, the Earth would always show the same face to the Moon, i.e. the rotations and the orbital motions of these two bodies would be synchronised.[3] How many times larger than at present would an Earth day and the Earth–Moon distance be with such perfect synchrony?

Assume that the orbit of the Moon remains circular, and neglect the tidal effect of the Sun.

P39** An astronaut, who seems light-headed, but who is an expert in celestial mechanics, jumps away from the International Space Station (ISS) in the direction directly opposed to that of the Earth, with a speed of $v_0 = 0.1\ \mathrm{m\ s^{-1}}$. He has an oxygen cylinder, but no lifeline tether or auxiliary jet pack. What will be his greatest subsequent separation from the ISS? If it is finite, for how long must his oxygen supply last?

P40* For an interstellar space mission, set in the future, an attempt is made to partially compensate for the lack of gravity by uniformly rotating the reasonably long and very heavy cylindrical space ship, which has a diameter of $2R = 20$ m, around its symmetry axis. The period of rotation is adjusted so that the astronauts feel an Earth-like 'gravitational acceleration' of $g = 10.00\ \mathrm{m\ s^{-2}}$ at the outer edges of the cylinder.

During the long journey, the astronauts exercise in a 5 m 'high' gym, whose 'floor' is the outer casing of the space ship; they notice that things are not quite like they are on Earth.

a) How much work is done by a (point-like!) astronaut with a mass of 80 kg, when he climbs up to the gym's ceiling on a fixed climbing pole. By how much, measured in an inertial frame of reference fixed to the space ship's centre of mass, does the astronaut's kinetic energy change as a result of the climb? How can the work done be reconciled with the change in his kinetic energy?

b) If the astronaut fell from the top of the pole, how long would he take to fall, and how far from the bottom of the climbing pole (measured along the floor) would he 'hit the ground'?

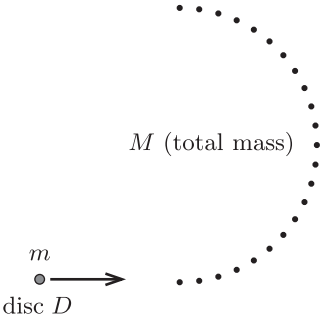
P41 The height difference between the top and bottom of a downward-moving escalator is $h = 20$ m. A mischievous boy of mass $m = 50$ kg runs up from the

[3] In reality, before this could happen, the Sun will expand into a red giant and engulf the current orbit of the Earth.

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bottom to the top at an (average) speed, relative to the steps, that is one-and-a-half times their translational speed. Find the work done by the boy and explain how it is accounted for.

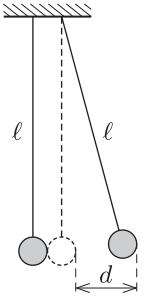
P42* On an air-hockey table, there are N identical small discs lying equally spaced around a semicircle (*see* figure); the total mass of the discs is M. Another small disc, D, of mass m, travelling in a direction perpendicular to the closing diameter of the semicircle, strikes the first of the stationary discs. By some miracle, it subsequently bounces off all of the other $N - 1$ discs in turn, after which it is travelling in a direction directly opposed to that of its initial motion. All the collisions are perfectly elastic, and friction is everywhere negligible.



a) In the limiting case of $N \to \infty$, what is the minimal value of the mass ratio M/m for such a miracle to be possible?

b) When the mass ratio has the critical value found in part *a*), what is the ratio of the final and initial speeds of D?

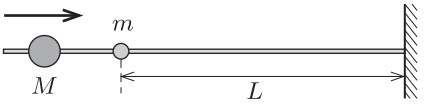
P43 Two identical balls are suspended on (vertical) threads of length ℓ so as to just touch each other and, as shown in the figure, can swing in the plane defined by their suspension points and their centres. One of the balls is drawn aside in this plane through a distance d ($\ll \ell$) and then released. Each time the balls collide, they do so inelastically, and, as measured in their centre-of-mass frame, their velocities decrease by a factor of k, the coefficient of restitution, where $0 < k < 1$.



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How will the balls be moving at a much later time? What will be the amplitude of the swings after a large number of collisions have taken place? Assume that the damping due to air resistance is very small.

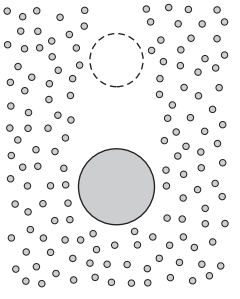
P44** Two small balls are threaded onto a frictionless horizontal rod that protrudes from a vertical wall. The lighter ball with mass m is initially at rest, a distance L from the wall, while the much heavier second ball, of mass M, approaches the wall from a distance greater than L (*see* figure). After their elastic collision, the ball of mass m slides towards the wall, bounces back elastically, and again collides with the heavier ball. The process then repeats itself, over and over.



How close does the heavier ball get to the wall?

Consider the balls as point-like particles, and, where appropriate, assume that $m \ll M$.

P45* In a very dense fog, there are many tiny water drops that 'float' in the air with negligible speed. If one of the water drops, which is a little larger than the rest, begins to sink, it absorbs those smaller drops that lie in its path (*see* figure). The ever-growing drop, which can be regarded as spherical, is found to be accelerating uniformly, despite the air drag – proportional to the square of the speed and the cross-sectional area of the drop – acting upon it. What is the maximum possible value for this acceleration?

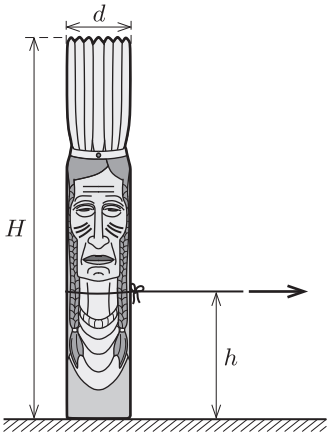


P46 A carved wooden American Indian statuette, which can be considered (to a good approximation) as a solid cylinder with a homogeneous mass distribution, has a height of $H = 6$ cm and a diameter of $d = 1$ cm (*see* figure). A thread is attached to the statuette at a point that is $h = 2$ cm above the base, and the statuette is placed in the middle of a rough horizontal table-top. The coefficient of friction between the table-top and the statuette is $\mu_{\text{kinetic}} = \mu_{\text{static}} = 1/3$.

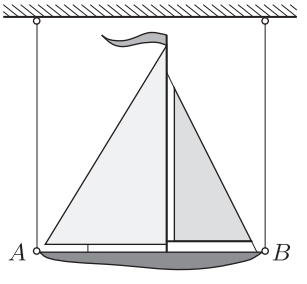
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As a challenge, contestants are required to pull horizontally on the thread and drag the statuette (in one continuous movement) to the edge of the table, without it falling over.

Is it possible to do this? If yes, how? If not, why not?



P47* In the shop window of a toy store, a miniature sailing boat is suspended (with its deck horizontal) by two vertical thin rubber bands (*see* figure). The masses of the mast and sails are negligible compared to that of the boat's hull, the length of which, from bow to stern, is much greater than its vertical height, from keel to gunwale.

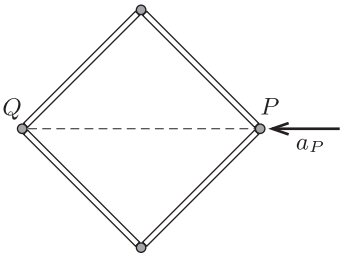


If the left-hand rubber band is cut, does point *B* sink or rise immediately afterwards?

P48* Four identical, homogeneous rods are connected by four light frictionless knuckle joints[4] to form a square, which is then placed on a horizontal, smooth, polished table-top. Vertex *P* is pushed in the direction of the diagonal of the square that passes through *P*, and, as a result, acquires an initial acceleration of a_P (*see* figure).

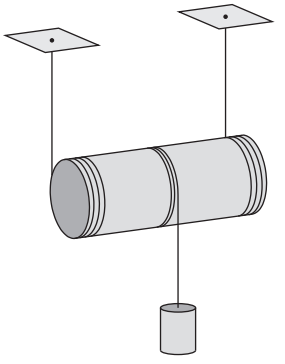
[4] A knuckle joint allows the angle between the two components it joins to vary freely in one particular plane.

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In which direction, and with how much acceleration, does the opposite vertex Q of the square start to move?

P49 A cylindrically symmetric (but not necessarily homogeneous) body is attached to two identical cords at points near its ends. The cords are partially wound in the same sense around the cylinder, and their free ends are fastened to points on a ceiling; initially, the cords are vertical and the cylinder is horizontal. A third cord is attached to and wound (in the same sense as the other two cords) around the middle of the cylinder; a heavy weight is tied to the free end of this cord (*see* figure).



When the system is released from rest, what is the acceleration of the heavy weight?

P50** A homogeneous flat disc (such as an ice-hockey puck) is both sliding and rotating on an icy surface. Because of friction, both kinds of motion decelerate and finally stop. Which of the two motions stops the earlier, the rotation or the translation?

Assume that the disc presses uniformly on the ice, that the frictional force between two surfaces does not depend upon their relative speed, and that air drag is negligible.

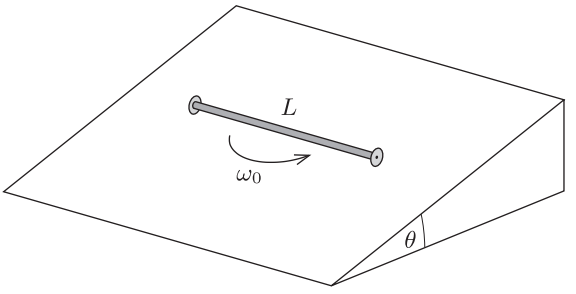
P51* A uniform rod of mass m and length L is fitted at each end with a frictionless bearing in the form of a freely rotating wheel, for which the rod acts

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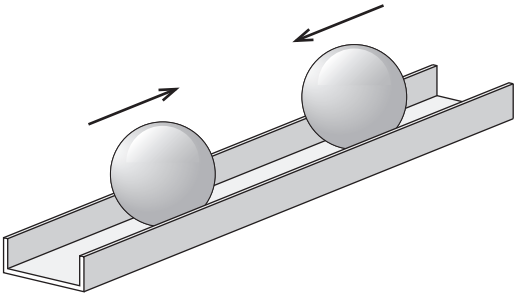
as an axle. The two bearings are identical and have negligible masses compared to that of the rod.

a) How does the rod move if it is placed on a horizontal rough surface – meaning that the bearings roll on it without slipping – and the two ends of the rod are initially given parallel velocities of v_1 and v_2 in a direction that is perpendicular to the axis of the rod?

b) How does the rod move if it is initially placed, with an angular velocity of ω_0 and zero centre-of-mass velocity, across the slope of a broad rough plane that is inclined at an angle of θ to the horizontal (*see* figure)?



P52 Two identical billiard balls of diameter 5 cm, each moving with a speed of 3 m s^{-1}, roll, without slipping, towards each other on a horizontal rectangular U-shaped trough that is sufficiently deep that the balls are clear of its base (*see* figure). The resulting instantaneous collision is perfectly elastic, and, during it, each ball reverses its linear velocity, though their angular velocities are not affected.



a) How wide does the trough need to be for the balls to collide twice?

b) Find the speed of the balls just before the second collision if the width of the trough is 4 cm.

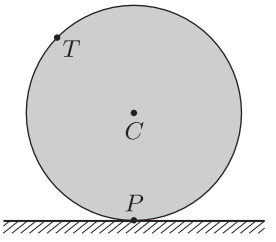
P53* A billiard ball, initially at rest on a billiard table, is struck by a cue tip at the point T shown in the figure. The cue lies in the vertical plane containing T, the centre C of the ball and the ball's point of contact P with the table;

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consequently, so does the line of action of the resulting impulse. Find the direction in which the cue should be aligned in order that, after the shot,

a) the ball's subsequent rotational and slipping motions terminate at the same instant, and the ball comes to a halt,

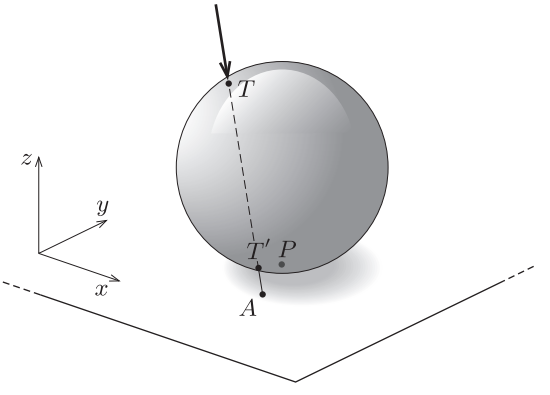
b) the ball rolls without slipping, whatever the value of the coefficient of static friction between it and the table.



Assume that, as a result of chalking the cue tip, the coefficient of friction between it and the ball is sufficiently large that there is no slippage between them during the cue stroke.

P54** If the line of action of the impulse in the previous problem does not lie in the vertical plane defined by the points T, C and P, then, just after the shot, the ball's angular velocity vector will not be perpendicular to the velocity of its centre of mass. Billiards players call this shot a *Coriolis-massé*.

Such a shot is shown in the figure, in which the line of action of the impulse meets the ball's surface (for a second time) at T' and the table at A.



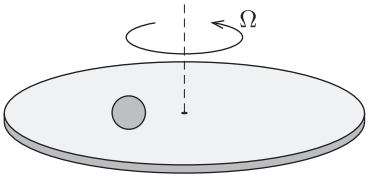
a) What kind of trajectory does the ball's centre of mass follow from just after the shot until the point at which simultaneous rolling and slipping cease?

b) In which direction, relative to the line PA, will the ball continue its path once it starts to roll without slipping?

Assume that, whatever the downward force acting on it, the billiard cloth does not 'become squashed', and the ball's contact with it is always a point contact.

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P55** A very large flat horizontal disc with a rough surface is rotating about its vertical axis of symmetry with angular velocity Ω. A solid rubber ball of radius R is placed on the disc in such a way that it rolls without slipping, and its centre moves in a circle that has radius r_0 and is concentric with the disc.

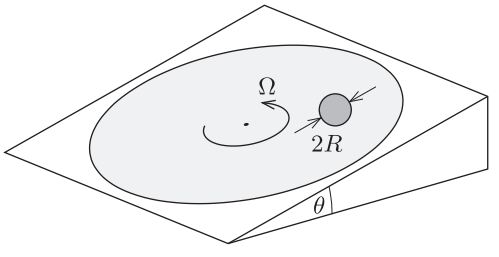


a) Find the initial velocity and angular velocity that the ball must have for this kind of motion.

b) How would the centre of the ball move if it were started from the same position with the same magnitude of initial velocity as specified in part *a*), but with the opposite direction?

P56* A large flat disc with a rough surface rotates around the axis of symmetry that is normal to its plane, and does so with constant angular velocity Ω. The plane of the disc is tilted at an angle θ relative to the horizontal. A magician places a solid rubber ball of radius R and mass m on the rotating disc and starts it off in an appropriate direction. Then, to the audience's great surprise, the centre of the ball moves uniformly in a straight line, until it reaches the rim of the rotating disc. Throughout the ball's motion, it does not slip on the disc, and the angular velocity of the disc does not change.

Find a physical explanation for this strange phenomenon. In which direction, and how quickly, should the magician start the ball for this stunt to be successful?



P57** A small solid rubber ball of radius r is thrown onto the inner wall of a long cylindrical tube, which has radius R and is fixed with its axis of symmetry vertical. If the ball is started off with a sufficiently large horizontal velocity v_0, then it starts to oscillate periodically in the vertical direction, while still maintaining contact with the tube. Describe the 'dance' performed by the centre of the ball.

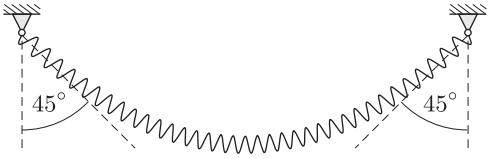
The static friction is quite large, and so the ball never slides on the wall. Assume that the ball is sufficiently incompressible that its contact with the tube

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is only ever through a single point, and that air drag and rolling friction are negligible.

P58 Slinky,[5] a well-known toy, is a pre-compressed helical spring, with a negligibly small unstressed length. To a good approximation, it obeys Hooke's law, but with a small 'Young's modulus', even being significantly stretched by its own weight.

Such a spring is first hung vertically with its upper end fixed, and is then re-suspended with its two ends attached to supports at the same vertical height, but separated horizontally. The ends of the Slinky make angles of 45° with the vertical, as shown in the figure. In which case is the stretched spring longer?



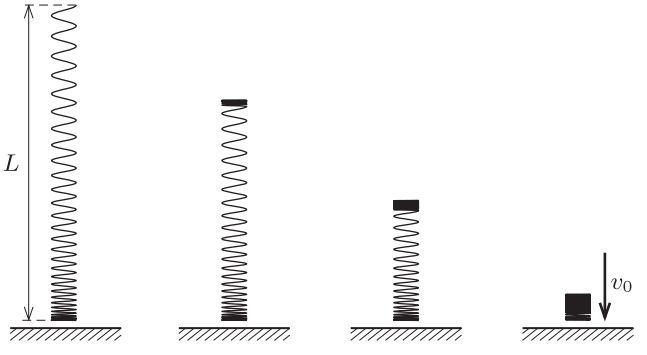
P59* What shape does a Slinky take up when its two ends are fixed to points that are at the same height and separated by a moderate distance (the Slinky remains a helical spring)?

P60* A Slinky of mass m was initially resting on a table-top with its axis vertical. Its top end was then slowly raised until its lowest coil was just clear of the table. At that point, the length of the Slinky was L.

a) How much work was done during the lifting stage?

b) If the upper end of the Slinky is now released (from rest), then, curiously, the lowermost coil does not start moving until the Slinky has completely collapsed (*see* figure). What speed, v_0, does the collapsed Slinky have immediately afterwards?

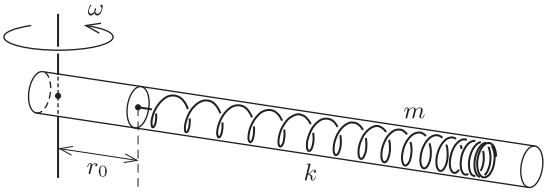
c) How long does the Slinky take to completely collapse?



[5] Slinky was invented by Richard James in the early 1940s; his wife dubbed the toy 'Slinky' (meaning 'sleek and graceful') after finding the word in a dictionary, and deciding that the word aptly described the sound of a metal spring expanding and collapsing.

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P61** A Slinky is placed inside a frictionless horizontal tube, with one of its ends attached to a fixed point of the tube. The fixed point is a distance r_0 from a vertical axis about which the tube rotates with uniform angular velocity ω (*see* figure). The Slinky spring is 'ideal': its unstressed length is negligible; its potential elongation is unlimited; and it obeys Hooke's law.



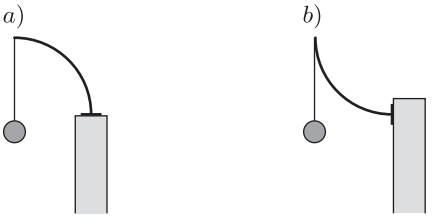
What is the length ℓ of the stretched spring, if its spring constant is k and its total mass is m?

Consider what happens in the limiting case $r_0 \to 0$.

P62** Find the shape of a Slinky inside the International Space Station (i.e. in weightless conditions) if it is rotating uniformly – like a skipping rope – with both ends of the spring twirled in unison.

P63* A tree in the editor's garden has a thin, light and elastic horizontal side branch that in early spring has very few leaves on it and provides a popular perch for wild birds. Is the end of the branch depressed more when a pigeon sits at its midpoint or when a blackbird, which only weighs one-quarter as much as the pigeon, perches on the end of the branch?

P64* A hat peg is planned by a post-modern interior designer in two versions. In both, a thin but strong elastic metal wire in the form of a quarter-circle is attached to a solid vertical post. The alternative mounting arrangements are shown in the figure.

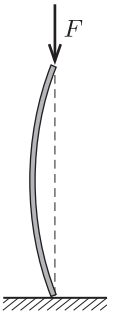


The designer is surprised to find that, when the hat pegs are loaded with the same weight (as shown), the ends of the pegs are not lowered by the same amount in the two designs. Find a simple argument that determines in which design the droop of the peg's tip is the larger.

P65* Hanging a body with a mass of 1 kg from the end of a uniform, horizontal, 1 m long rod, which is fixed rigidly at its other end, causes the loaded end to

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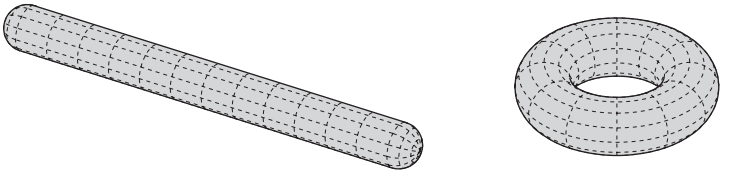
deflect by 1 cm. If the rod is stood vertically on one end (*see* figure), *estimate* the vertical load F, applied to the other end, that is required to make it buckle.



P66 A cable of given thickness, and large but finite tensile strength, is to hang vertically downwards from a ship, with the intention that it should reach the sea bed. However, when the lowered cable is 1 km long, it snaps (at the top) because of its own weight. How much cable is required to reach the sea bed, at a depth of 3 km, if filaments can be doubled up and run parallel to each other along particular lengths? Estimate the total length of original cable needed to construct a composite one that will reach to the bottom of the Mariana Trench, which lies 11 km beneath the surface.

Assume that the density of seawater is constant, and that the stretching of the cables is negligible.

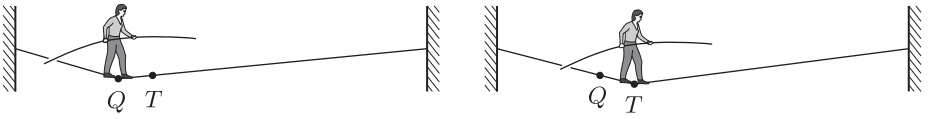
P67** It is a common observation in the kitchen that, if a frankfurter sausage (a long, straight sausage in ovine gut skin, sometimes called a 'Frankfurter Rindswurst') splits during the boiling process, it always does so 'lengthways', and never 'across'. What is the reason for this?



If it were possible to produce one, where, and along which direction, would a toroidal sausage split while being cooked? Assume that, in both cases, the thickness of the sausage skin is uniform.

P68* A thin, strong, but flexible steel tight-rope is installed horizontally above a wide street, and highly tensioned. An acrobat moves slowly onto and along the tight-rope. When he reaches the quarter-way point (Q in the figure), the nearer trisection point of the rope (T) has been depressed by 5 cm from its original position. By how much is Q depressed, when the acrobat reaches point T?

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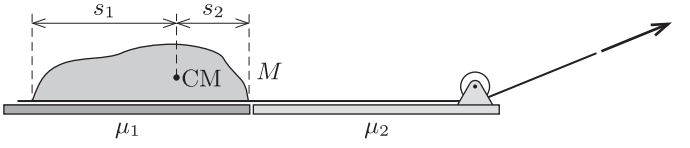
Can your result be generalised to arbitrary points Q and T?

You may assume that the weight of the rope is negligible, that its depression is always very small compared to its total length, and that the tension in it can be treated as constant.

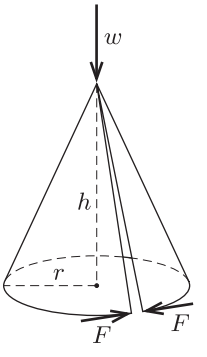
P69 Devise a mechanical system that uses a fundamental physics principle to determine the point P on a general triangle that has the property that the sum of its distances from the three vertices is a minimum. Can your device be adapted to minimise a *weighted* sum of the distances?

P70 A sack, full of sand and of mass M, lies on a carpet, which, in turn, lies on a relatively smooth horizontal surface. The distribution of the sack's weight is not uniform, but it is known that its centre of mass (CM) lies at distances s_1 and s_2 from its extremities.

By pulling horizontally on the carpet, both it and the sack are to be moved onto an immediately adjacent rougher surface at the same height (*see* figure). How much work is required to do this if the coefficients of friction between the carpet and the original and final surfaces are μ_1 and μ_2, respectively?



P71 A cone with height h and a base circle of radius r is formed from a sector-shaped sheet of paper. The sheet is of such a size and shape that its two straight edges almost touch on the sloping surface of the cone. In this state the cone is stress-free.



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The cone is placed on a horizontal, slippery table-top, and loaded at its apex with a vertical force of magnitude w, without collapsing. The splaying of the cone is opposed by a pair of forces of magnitude F acting tangentially at the join in the base circle (*see* figure). Ignoring any frictional or bending effects in the paper, find the value of F.

P72 Frank has made an iron triangle by welding together three thin iron rods of identical cross-section. He decides to try to identify the centre of mass of the triangle, and so he lays it on a sheet of paper, and draws the lines connecting each vertex to the midpoint of the opposite side. At that moment, Lisa arrives and says that the centre of mass of the triangle is not at its centroid (its geometrical centre). She claims that Frank should connect the midpoints of the triangle's sides to form a smaller triangle, and that the centre of mass of the larger iron triangle coincides with the *incentre*[6] of the smaller one. Who, if anybody, is right?

P73 A triangle is cut out from a uniform sheet of cardboard, and placed on a level table-top standing on one of its edges. It is loosely supported on both sides so that it can move only in its own plane – but can do so freely in that plane. Is it possible to make a triangle that tumbles over from two of its edges, and has only a single equilibrium position?

P74** We describe any particular face of a tetrahedron as 'unstable' or 'stable' according to whether or not the tetrahedron spontaneously falls over if it is placed on a level table-top with that face as its base. Is it possible to make a (homogeneous) tetrahedron that has three unstable faces and only a single stable one?

P75 The cable strung between two neighbouring electricity posts sags a little. The mass per unit length of the cable is λ, the distance between the posts is L, and the 'maximum sag' of the cable is d ($d \ll L$). What is the (approximate) tension in it?

P76* In a gymnasium, a climbing rope and a climbing pole, both with uniform cross-sections, have the same length; they also have equal weights. Each is attached to the ceiling of the gymnasium by a small pivot, and both are pulled aside by identical horizontal forces at their lower ends. Determine whether it is the rope's or the climbing pole's lowest point that is lifted higher.

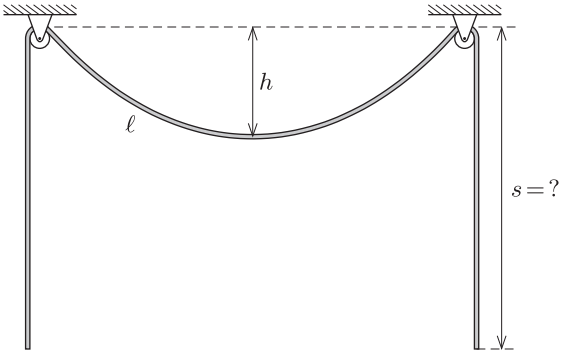
P77 The two ends of a 40 cm long chain are fixed at the same height, as shown in the figure. Find the radius of curvature of the chain *a*) at its lowest point and *b*) at the suspension points.

[6] The incentre of a general triangle is the (unique) point at which the internal bisectors of its angles meet; this point is the centre of the triangle's incircle.

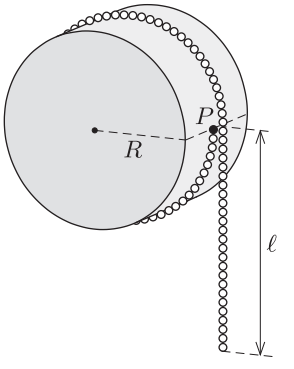
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P78** A uniform flexible rope passes over two small frictionless pulleys mounted at the same height (*see* figure). The length of rope between the pulleys is ℓ, and its 'sag' is h. In equilibrium, what is the length s of the rope segments that hang down on either side?

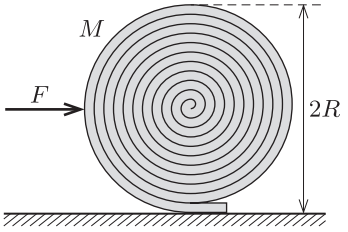


P79* One end of a necklace of small pearls is attached to the outer surface of a fixed cylinder that has radius R and a horizontal axis; the attachment point P is at the same level as the axis. The necklace is wound once round the slippery surface of the cylinder, and the free end is left to dangle (*see* figure). How long, ℓ, does this free end need to be if the rest of the necklace is to touch the cylinder surface everywhere?



P80* A flexible scatter rug (also known as a runner), of mass M and length L, is tightly coiled into a cylinder of radius R ($\ll L$), as in the figure. If the coiled rug is released, then, in the absence of rolling friction, it spontaneously unrolls to its full length.

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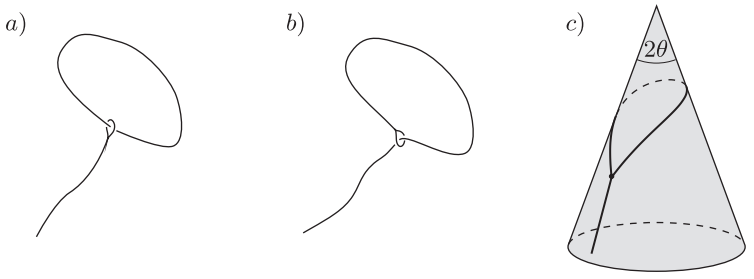


a) Explain in terms of the forces involved why this happens.

b) How large a horizontal force F, applied as in the figure, is required to prevent the unrolling of the rug?

P81** The committee of the Glacier Climbing Club has decided to introduce a new challenge for its members: they have to climb as high as they can on an artificial right-circular cone, the surface of which has been made into a very smooth, slippery 'iceberg', by letting water trickle down it in sub-zero temperatures. A lasso is to be their only piece of climbing equipment!

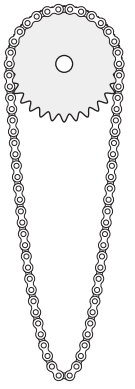
The lasso for novices is as shown in figure *a*) and consists of a length of rope attached by a small eyelet to a closed loop of *fixed* length. That for experts is shown in figure *b*) and is a single rope with, at one end, an eyelet, through which the other end is threaded. All the ropes are light compared to the mass of a climber, and friction between them and the ice, and within the eyelet, is negligible.



For what ranges of cone angle, 2θ, are (i) the novices and (ii) the experts able to climb up the iceberg using their lassos in the way illustrated in figure *c*)? Assume that the straight segment of a lasso follows one of the cone's lines of steepest descent.

P82* The spindle of a bicycle chain assembly is mounted horizontally, and a loop of bicycle chain is placed on the toothed wheel, as in the figure. The wheel is then rotated around its axis at a steadily increasing rate until it has achieved a high, but constant, angular velocity. What is then the shape of the chain?

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P83** A fire hose of mass M and length L is coiled into a roll of radius R ($\ll L$). The hose is sent rolling across level ground with an initial speed v_0 (and angular velocity v_0/R), while the free end of the hose is held at a fixed point on the ground (*see* Fig. 1). The hose unrolls and becomes straight.

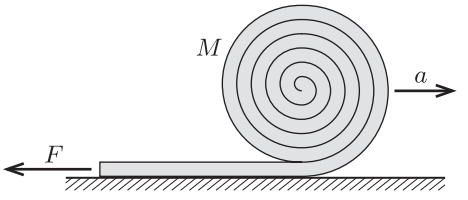


Fig. 1

Peter and Pauline (two students studying physics) are discussing what happens during the unrolling. They agree that in some respects simplification of the analysis can be justified: that, if the initial kinetic energy of the roll is much greater than its potential energy ($v_0 \gg \sqrt{gR}$), then the effect of gravity can be neglected; that the hose can be considered as arbitrarily flexible; and that the work necessary to deform the hose, and to overcome both air drag and rolling friction, can be neglected. However, they do think that it is important to investigate the roll's vertical motion in addition to the more obvious horizontal one.

The speed of the diminishing roll continually increases, and its acceleration a is clearly a vector pointing in the same direction as its velocity. On the other hand, the vector resultant of the horizontal external forces that act on the roll (the frictional force and the restraining force at the fixed end of the hose) points in the opposite direction. This strange fact can be understood by recognising that the total momentum of the moving roll (and hence of the whole system) is

$$p(x) = m(x)\, v(x) = M(1 - (x/L))\frac{v_0}{\sqrt{1 - (x/L)}} = Mv_0\sqrt{1 - (x/L)},$$

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where $m(x) = M(1 - (x/L))$ is the mass of the moving part of the hose after it has travelled a distance x. Its speed $v(x)$ has been determined using the conservation of mechanical energy (but ignoring gravitational effects), and is given by $v(x) = v_0/\sqrt{1 - (x/L)}$.

Clearly, as x increases, $p(x)$ decreases – reflecting the fact that the mass of the piece in motion decreases faster than the rate at which its speed increases. The direction of the resultant force $F(x)$ acting on the system is therefore opposed to that of the motion,[7] with

$$F(x) = \frac{\mathrm{d}p}{\mathrm{d}t} = \frac{\mathrm{d}p}{\mathrm{d}x}\frac{\mathrm{d}x}{\mathrm{d}t} = \frac{\mathrm{d}p(x)}{\mathrm{d}x}v(x) = -\frac{Mv_0^2}{L}\frac{1}{2(1 - (x/L))}.$$

Peter (using the conservation of mechanical energy and mass) has determined not only the mass $m(x)$ of the roll, its radius $r(x)$ and its angular velocity $\omega(x)$, but also its moment of inertia $I(x)$. He has also found the horizontal and vertical velocities, v_x and v_y, of the roll's centre of mass (*see* Fig. 2). All of these quantities are functions of x, and therefore also of time t.

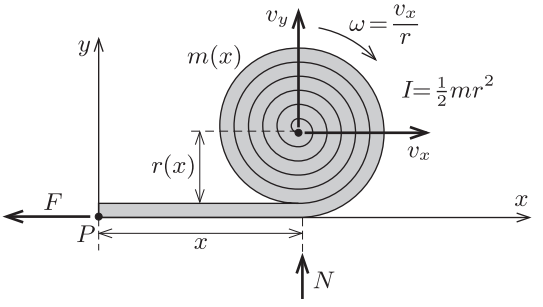


Fig. 2

Peter says that:

> I have found v_x and v_y, and multiplying them by the instantaneous mass $m(x)$, I have calculated the components p_x and p_y of the linear momentum of the roll, and from the rates of change of the latter, the external forces, F and N, can be found.

He decided to calculate the angular momentum J of this strange system, and also its rate of change, and so check whether or not the torques due to the external forces would produce this rate of change, i.e. whether or not $\tau_{\text{net}} = \mathrm{d}J/\mathrm{d}t$. As both the centre of the roll and the centre of mass of the whole system are accelerating, he expressed the formulae for the angular momentum and the torques, not about these points, but with respect to the fixed point P, the stationary end of the hose.

[7] This calculation appears in the predecessor of this book, as the solution to 'Problem 108' of P. Gnädig, G. Honyek & K. F. Riley, *200 Puzzling Physics Problems* (Cambridge University Press, 2001). There, a further question was posed: How long does it take for the hose to completely unroll?

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To his great surprise, he has found that – though the net torque of the external forces is *zero* – the angular momentum of the system does not remain constant, but *changes with time*! After he had checked his calculations for the umpteenth time, he was close to stating:

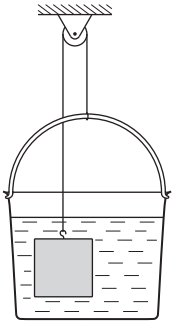
> In this system, the angular-momentum theorem, a basic law in classical mechanics, is not obeyed.

Pauline, however, thought differently! She claimed that there was an error in the method Peter had used, in that he had applied a well-known standard formula to a situation to which it did *not* apply.

Who was right?

P84* A cylindrically shaped closed container rotates uniformly around its principal symmetry axis, which lies horizontally, at a rate of 0.5 revolutions per second. Both the cylinder's inner diameter and its length measure 1 m, its inner wall is rough and it contains 100 kg of sand. Estimate the temperature rise of the sand during 10 minutes of operation. Make realistic estimates for (nearly all of) the data required, and neglect heat loss through the wall of the container.

P85 A solid iron cube of volume 10^{-3} m^3 is fastened to one end of a cord, the other end of which is attached to a light plastic bucket containing water. The cord, which has negligible mass, passes over a pulley, and the iron cube is suspended in the water, as shown in the figure. It is found that the system is in equilibrium.



a) How many litres of water are there in the bucket?

b) What would happen if more water were poured into the bucket?

c) What would happen if some or all of the water evaporated?

P86 An air bubble with a diameter of a few centimetres was injected into the bottom of a giant *closed* vertical cylinder that had a height of 10 m and was, apart from the bubble, *completely* full of water. At the moment of injection, the water pressure at the bottom of the cylinder was 1.1 atm. The air bubble rose, and a little

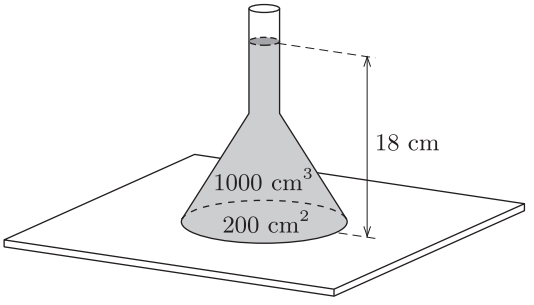
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bit later it reached the top of the closed container. Find the pressure of the water at the bottom of the cylinder after the air bubble had risen.

For the sake of definiteness, assume that the hydrostatic pressure of 10 m of water is exactly 1 atm.

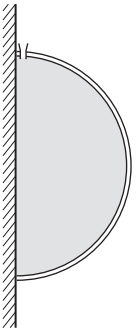
P87 There is water in a container that stands on a table. A rubber hose connects the bottom of the container with a cupboard-sized *black box* standing next to the table. If additional water is added to the container, then the water level in it *sinks*, and if some is taken out, the level rises. Whatever could the black box contain?

P88 A funnel is placed upside down on a table, as shown in the figure, and 1000 cm^3 of water is poured into it. The area covered by the funnel on the tabletop is 200 cm^2, and the height of the water is 18 cm.



What is the minimum mass of the funnel if it is not to be lifted away from the supporting surface?

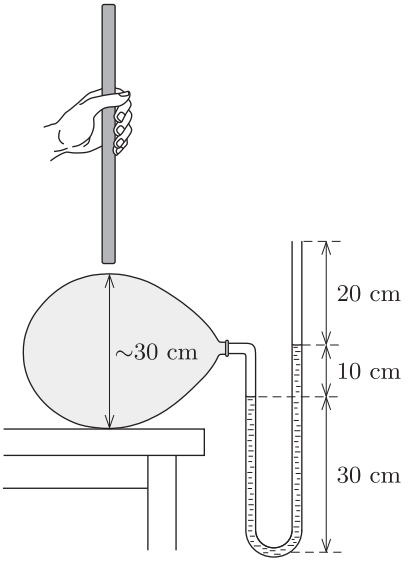
P89* A thin-walled hemispherical shell of mass m and radius R is pressed against a smooth vertical wall, and, through a small aperture at its top, is filled with a liquid (say, water) of density ϱ (*see* figure). What are the minimal magnitude and the direction of the force that has to be applied to the shell if liquid is not to escape from it? To which point on the shell should this force be applied?



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P90 The gauge pressure[8] in the tyres of quite an old car (one without power steering) is increased from 1.5 atm to 2.0 atm. What percentage change is there in the force required to rotate its steering wheel when the car is stationary?

P91* A balloon is inflated and connected to a water manometer, as shown in the figure. The difference in height between the water columns in the two arms of the manometer is 10 cm. When a solid cylindrical iron rod, of length 50 cm and diameter 2 cm, is placed vertically and carefully on top of the balloon, will the water in the manometer overflow?



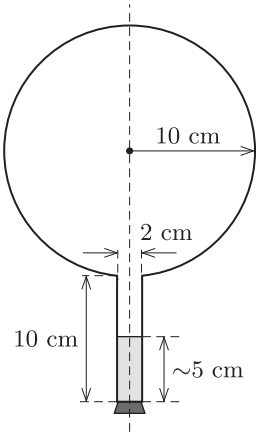
P92 The wall of a vertical measuring cylinder contains many uniformly distributed small holes. The cylinder is filled with water up to a height H, and so thin water jets are ejected horizontally through the holes. The jets do not interfere with each other, and the water level in the cylinder is continuously maintained at its initial value. What shape is the envelope of the water jets?

P93** There is some water in a closed spherical Florence flask.[9] When the flask is turned upside down, the water collects in its long neck up to a height of about 5 cm. The internal dimensions of the flask are shown in the figure.

[8] The gauge pressure is not the absolute pressure, but the difference between the absolute pressure and the atmospheric pressure p_0.

[9] A Florence flask is a spherical flask (sometimes with a flattened bottom) that is mainly characterised by its long (and sometimes quite wide) neck.

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The flask is now rotated about its vertical axis of symmetry at a rate of three revolutions per second, and it is arranged that the temperature of the flask wall is everywhere the same. After a sufficiently long time, an equilibrium state is established in the flask.

Make a sketch showing the distribution of the water within the flask when this equilibrium state has been reached.

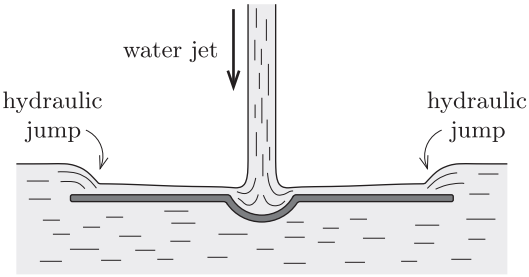
P94* A razor-blade floats on the surface of water contained in a glass. When the glass is gently shaken, the razor-blade sinks. How, if at all, does the water level change as a result?

P95* Take a disc of copper with a diameter of about 10 cm, and a thickness of approximately 0.2 mm (other metals, such as soft steel or aluminium, are also suitable for this experiment). In the centre of the disc, make a circular depression with a diameter of 15–20 mm and a depth of 2–3 mm. If such a disc is placed on a water surface, it will probably float (*see* solution on page 263). But if some water is splashed onto it, the disc will almost certainly sink, because metal is denser than water.

Now place the dry disc on the water surface and direct a strong vertical jet of water downwards into the central depression. What you will observe is paradoxical – the jet pushes the disc down, but cannot sink it![10] It will be noticed that, on the disc's surface, there is a circular 'hump' of water that is pushed outwards by the diverging thin layer of running water (*see* figure).

[10] This interesting phenomenon is described in an article entitled 'An unsinkable disk' by A. Luzin in *Quantum*, Sept./Oct. 1999, p. 42. Unfortunately, the explanation given there of why the disc does not sink is, in our opinion, in error.

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In hydraulics, the observed water 'hump' is known as a 'hydraulic jump', and is characterised by a sharp, step-like rise of the water level in an open waterway; at the jump, the nature of the water motion changes. When liquid moving at high speed discharges into a zone in which it has a lower velocity, a rather abrupt rise occurs in the liquid level: the rapidly flowing liquid is suddenly slowed and increases its depth.

A common example of a hydraulic jump is the roughly circular stationary wave that forms around a central stream of water falling on the flat bottom of a kitchen sink. The 'hump' of water near the rim of the copper disc is very similar to this easily observed phenomenon. The depression at the centre of the disc provides the disc with horizontal stability – without it, the disc would move rapidly away from under the water jet.

Now for the question! Why does the disc not sink under the pressure generated by the powerful water jet?

P96 Two spherical soap bubbles with different initial radii coagulate, and the radii of their free surfaces after the coagulation are R and r. What is the radius of curvature of the soap film that separates the two bubbles? What is the radius of this film's circular perimeter?

P97* While cleaning his flat, a physics research student knocks over a bucket containing 5 litres of water, which then forms itself into one continuous puddle. How large, in area, does the student expect the puddle to be? The contact angle between the floor and the water is 60°.

P98* Inside the cabin of a freely orbiting space ship, a ball of water approximately 4 cm in diameter and a nearby thin glass rod are in a state of levitation. The rod is about 8 cm in length, and has a circular cross-section and blunt ends. One end of the rod gently touches the ball. Sketch the shape the 'water drop' subsequently takes up.

P99* Inside a space station, a freely floating, closed, spherical shell of inner diameter 8 cm has one-third of its volume occupied by water; the rest contains

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air at STP (standard temperature and pressure). In equilibrium, how is the water distributed, if the material of the spherical shell is

a) glass, which is 'perfectly' wetted by the water,

b) silver, for which the contact angle of water is 90°?

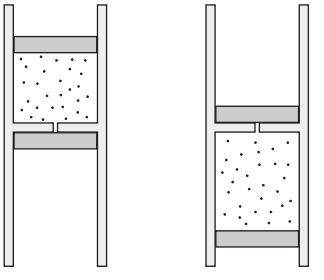
P100* You are given 1 kg of distilled water at 0 °C, an equal mass of boiled tap water at 100 °C, and the task of warming the distilled water to 60 °C. How would you set about it?

No additional water, hot or cold, is available, but you are provided with both insulating and conducting materials (in sheet form) and a suitable selection of tools.

What is the maximum possible temperature, in principle, to which your method could warm the distilled water?

P101* There is a *very small* hole in an otherwise totally enclosed heated furnace. Outside the furnace, the air temperature is 0 °C and the air pressure is 100 kPa. The air inside the furnace is kept at a constant temperature of 57 °C by the controlled heating system, and after a sufficiently long time its pressure also becomes stationary. Estimate the magnitude of this stationary pressure.

P102* An ideal gas, enclosed in a fixed, long, vertical cylinder with open ends, is separated from its surroundings by two identical, frictionless pistons. Between the pistons there is a fixed, rough, separating wall in which there is a small hole (*see* figure). Initially, the temperature of the enclosed gas is equal to that of the outside air, and the lower piston is held up against the separating wall. When the lower piston is released, the pistons descend slowly.



In which of the two following scenarios will the bottom piston finish in a lower position:

a) the pistons, the separating wall and the wall of the cylinder are good heat insulators,

b) the pistons, the separating wall and the wall of the cylinder are good heat conductors?

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P103* It is well known to hikers and mountaineers that the air temperature decreases by 1 °C for each 100 m increase in altitude.

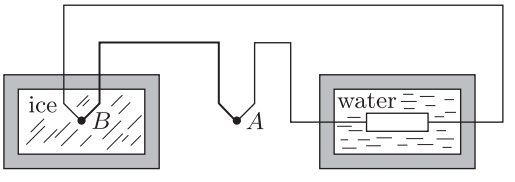
The ground, warmed by solar radiation, heats up the air above it, and the lowest layer of the atmosphere (the troposphere) is in a state of permanent convection. However, this mechanism is insufficient to equalise the temperatures of the various air layers, because rising air masses are not able to effectively exchange heat with their surroundings (due to a combination of their rapid motion and the low thermal conductivity of air).

There is thus a variation of temperature with height. Use simple physical considerations to explain and estimate the observed rate of air temperature decrease with altitude.

P104 A quantity, n mol say, of helium gas undergoes a thermodynamic process (neither isothermal nor isobaric) in which the molar heat capacity C of the gas can be described as a function of the absolute temperature by $C = 3RT/(4T_0)$, where R is the gas constant and T_0 is the initial temperature of the helium gas. Find the work done on the system up until the point at which the helium gas reaches its minimal volume.

P105* An ideal gas is contained within a closed bag made from material that is both easily stretched and an excellent heat insulator. The pressure outside the bag is decreased from p_1 to p_2 and the gas both expands and cools. In which case is the temperature drop in the gas greater: if the decrease in the outside pressure is slow, or if it happens suddenly?

P106** One of the junctions (A) of a thermocouple is in air at $T_A = 27\,^{\circ}\mathrm{C}$, while the other ($B$) is placed inside an insulated vessel containing ice at $T_B = 0\,^{\circ}\mathrm{C}$. The electrical energy produced by the thermocouple is dissipated in a resistor of resistance R placed in an insulated water bath. The masses of the water and ice are equal. Find the amount by which the temperature of the water has increased when the last of the ice has melted.



P107** By how much can a small spherical black body be warmed by sunshine, using a thin convex lens that has a focal length equal to twice its diameter? Does the result depend upon the radius of the sphere?

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P108 During cloudless autumn nights, hoar frost can cover car windscreens, even if the temperature of the surrounding air does not fall below 0 °C. How can this happen? Where should a car be parked so as to avoid this phenomenon?

P109* The three vertices of a metal plate in the form of an equilateral triangle are held at constant temperatures T_1, T_2 and T_3 (by cooling or heating, as necessary). What temperature do you expect at the centre of the triangle? Prove that your answer is correct.

P110 Identical ordinary ice cubes are put into two quite large beakers, one of which contains tap water, the other a strong brine solution. The liquids have the same volume, and both of them are at room temperature. In which beaker will the ice cube melt more quickly?

P111 If 1 kg of water freezes, then 334 kJ of energy are released. How much energy is released if 1 kg of supercooled water at −10 °C freezes, and during the freezing process the temperature remains constant?

P112 Should we increase or decrease the volume of a quantity of air that contains saturated water vapour, if we wish to condense some of the vapour?

P113* Water is boiling in a narrow test tube that is open at the top. Just before the last few drops are vaporised, the tube is rapidly and hermetically sealed. The temperature at the top of the tube is then slowly increased to 200 °C, while the very bottom of it is maintained at 100 °C.

What is the pressure of the steam in the test tube?

P114* At some time in the distant future, humankind makes contact with the inhabitants of an exoplanet, on which the atmospheric pressure near the surface is the same as on the Earth, i.e. 1 atm ≈ 101 kPa. Further, the atmosphere consists of a mixture of oxygen and nitrogen gases. Because of these similarities, the planet is called Exo-Earth.

Human researchers and Exo-Earth scientists cross-check the physical and chemical data of their two atmospheres, and state that, on both planets, the boiling points of liquid nitrogen and liquid oxygen are 77.4 K and 90.2 K, respectively, at standard atmospheric pressure.

On both planets, local 'air' was isothermally compressed at a constant temperature of 77.4 K, and liquefaction set in when the pressure reached 113 kPa. However, on Earth *oxygen*, and on Exo-Earth *nitrogen*, condensed first.

a) What is the composition of the atmosphere on Exo-Earth?

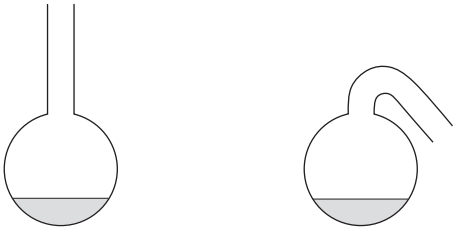
b) For what atmospheric composition would the oxygen and nitrogen begin to liquefy *simultaneously* under isothermal compression at 77.4 K, and at what pressure would this happen?

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P115 Two spherical flasks are almost identical. One of them has a straight neck directed upwards; the other is similar to a retort, in that its neck points downwards, as shown in the figure. The same amount of a liquid is put into each of the flasks, which are then heated from below in a controlled way that ensures that the temperatures of the liquids within them remain fixed and identical. Which flask will run out of liquid first if

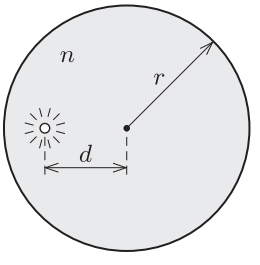
a) water is used, and the constant temperature is 100 °C,

b) ether (diethyl ether, $(C_2H_5)_2O$) is used, and the constant temperature is 34.6 °C, the boiling point of ether at atmospheric pressure?



P116* Two narrow, straight, vertical test tubes with identical cross-sectional areas are open at the top; one of them is 20 cm long, and the other is 40 cm. We pour 1 cm^3 of eau-de-Cologne into the first test tube, and 2 cm^3 into the second. How many times longer does the latter take to evaporate than the former? Is there any change in the result if both test tubes are covered, and identical but very small holes are made, one in each cover?

P117** A point source of light is inside a solid sphere made from glass of refractive index n. The sphere has radius r and the source is a distance d from the sphere's centre (*see* figure). What is curious is the fact that the sphere forms a *perfect* (virtual) image of the light source, i.e. the backward continuations of all light rays starting from the source, refracting at the sphere's surface, and travelling to the right-hand side of the figure, intersect each other at a common point.



What is the distance d, and where is the sharp image of the light source formed?

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P118** What is the shape of the blunt end of a glass rod that would focus (at a single point F within the rod) all the light rays incident upon it that are parallel to the rod's axis? Find the equation describing the shape of the surface, expressing it in terms of the (uniform) refractive index n of the glass, and the rod's focal length f, defined to be the distance between F and the point on the surface that lies on the rod's axis.

What is the corresponding solution if a beam propagating inside the glass rod, and parallel to its axis, is to be brought to a focus at an axial point outside the rod?

P119 On a spherical planet, the refractive index of the atmosphere, as a function of altitude h above the surface, varies according to the formula

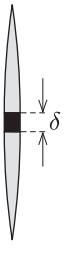
$$n(h) = \frac{n_0}{1 + \varepsilon h},$$

where n_0 and ε are constants. Curiously, any laser beam, directed horizontally, but at an arbitrary altitude, follows a trajectory that circles the planet. What is the radius of the planet?

P120* Find the shape of the image that is formed by a converging thin lens with focal length f, if the object is a sphere of radius r ($< f$), and the centre of the sphere is at the focus of the lens.

P121* How many times 'brighter' is an image of the Moon when looked at through a telescope rather than with the naked eye? And what about the stars?

P122* A converging lens of focal length f is cut along a plane that contains the optical axis of the lens, and a small, black plate of thickness δ is placed between the two half-lenses. A point-like source emitting monochromatic light of wavelength λ is located on the 'optical axis', a distance p from the lens ($p > f$).



How many interference fringes can be seen on a screen placed a distance H behind the lens, with its plane perpendicular to the optical axis?

Data: $f = 10$ cm, $p = 20$ cm, $\delta = 1$ mm, $\lambda = 0.5$ μm, $H = 50$ cm.

P123* In an unusual optical diffraction grating, the distances between the neighbouring slits are not equal; they are alternately d and $3d$. The widths of the

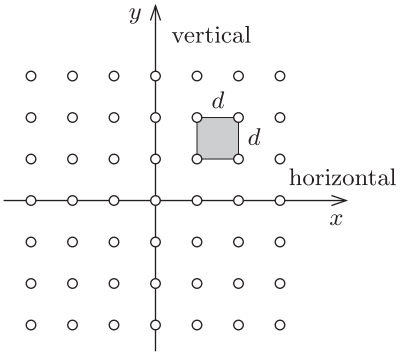
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slits themselves *are* all equal and are much smaller than d. If light of wavelength λ falls at normal incidence on the grating, what kind of diffraction pattern is formed on a screen placed at a distance L from it?

P124* An optical grating is illuminated normally by a laser beam of wavelength $\lambda \ll d$, where d is the slit spacing. The grating is unusual, in that alternate slits are wider and narrower: the width of the odd-numbered slits is a, while that of the even-numbered ones is b, where $b < a$ and both of them are much smaller than d.

The special character of this grating is reflected in its diffraction pattern in a peculiar, easily noticeable, way. How? What is the diffraction pattern like (i) if $b \ll a$ and (ii) if $b \approx a$, and the screen on which the patterns are formed is a distance L behind the grating?

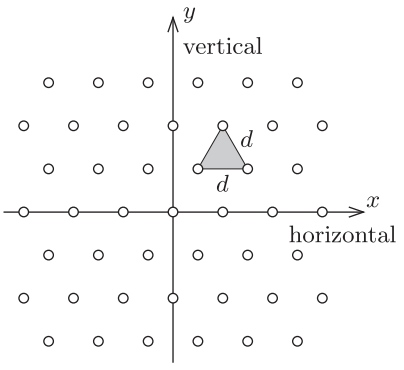
P125 An opaque sheet is perforated by many small holes arranged on a square grid (*see* figure), and is illuminated normally by monochromatic laser light of wavelength λ. What kind of diffraction pattern can be observed on a screen placed parallel to the sheet, and a distance L behind it, if the 'lattice constant' of the grid is d? Assume that $L \gg d \gg \lambda$.



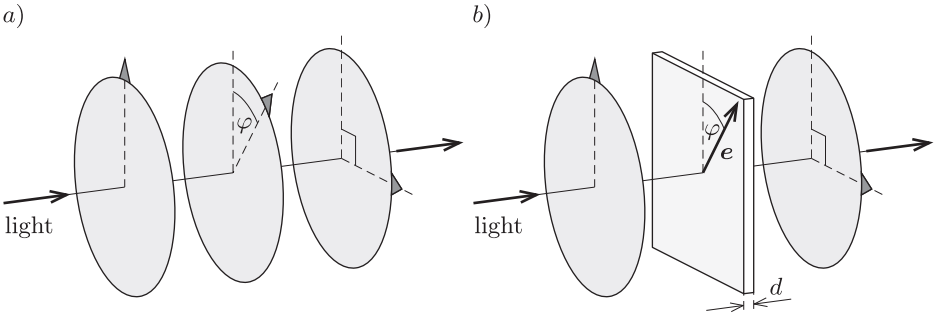
How does the diffraction pattern change if the sheet is compressed horizontally (along the x-axis in the figure) by a factor of N, with the result that the holes lie on a rectangular grid?

P126* An opaque sheet is perforated by many small holes arranged on a triangular grid (*see* figure), and is illuminated normally by monochromatic laser light of wavelength λ. What kind of diffraction pattern can be observed on a screen placed parallel to the sheet, and a distance L behind it, if the 'lattice constant' of the grid is d? Assume that $L \gg d \gg \lambda$.

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P127* *a*) Light cannot be transmitted through two coaxial polarising filters if their polarisation axes are orthogonally oriented. However, if a third filter is placed between them, some light can pass through all three. What is the maximum fraction of the incident intensity that can be so transmitted? Find the corresponding angle between the polarisation axis of the first filter and that of the one interposed – denoted by φ in figure *a*).



b) There is a second way in which light can be made to pass through the two orthogonal polarising filters: by placing, between and parallel to the filters, a uniform plate made from uniaxial birefringent (birefractive) material. The special feature of such a plate is that its refractive index for light polarised parallel to the direction of vector $\boldsymbol{e}$, which lies in the plane of the plate (*see* figure *b*)), is n_1, whereas for light polarised perpendicular to $\boldsymbol{e}$ it is n_2.

What is the maximum fraction of the incident intensity that can be transmitted, if the system is illuminated (perpendicularly to the plane of the filters and the plate) by monochromatic light of wavelength λ? Find the thickness d of the birefringent plate appropriate to this case, and how we should choose the orientation of $\boldsymbol{e}$?

P128** *a*) Nick, a 17-year-old grammar school pupil tidying up his room, found an old pair of spectacles used for viewing 3D movies. He tried on the specs in the bathroom, and he realised that when looking at his own image in the bathroom mirror, with one eye closed, he could see only his open eye; the other 'lens', in

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front of his closed eye, seemed totally dark. What can be the explanation for Nick's observation?

b) A week later, Nick went to a 3D multiplex cinema (movie theatre), and on returning home he repeated his experiment, but with the new 3D glasses he had just acquired at the cinema. But, to his great surprise, the effect was just the opposite of what he expected. In the mirror, he could see only his closed eye, and not the open one. Find the explanation for Nick's peculiar experience.

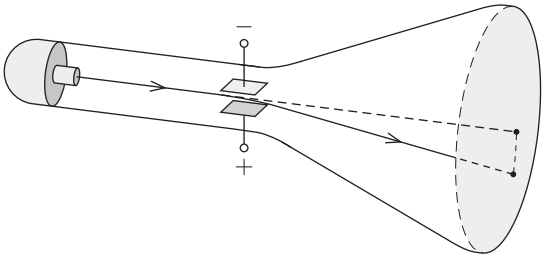
P129* A finite system of point electric charges, in vacuum and very far from anything else, is in stationary equilibrium under the influence of its own internal electric fields. What is the electrostatic interaction energy of the system? Is the equilibrium stable or unstable?

P130 Three small positively charged pearls lie one at each vertex of a triangle. Their masses are m_1, m_2 and m_3, and their charges are Q_1, Q_2 and Q_3, respectively. When the pearls are released from rest, each moves along a different straight line, the three motions taking place in a vacuum with negligible effects due to gravity.

What special condition has to be satisfied for this to happen? Find the angles of the triangle formed by the pearls at the beginning of their motion, if the charge-to-mass ratios of the three pearls are in the proportion

$$\frac{Q_1}{m_1} : \frac{Q_2}{m_2} : \frac{Q_3}{m_3} = 1 : 2 : 3.$$

P131* In a cathode ray tube, the beam emerging from the electron gun can be deflected by passing it through the electric field of a small parallel-plate capacitor. If the capacitor is uncharged, then the speed and direction of motion of the electrons are constant, and they hit the screen with an unchanged velocity.



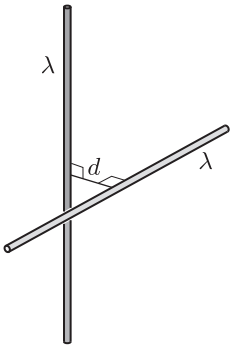
When the parallel plates of the capacitor carry a *constant charge*, the electron trajectory is bent. Is the speed with which electrons then hit the screen increased or decreased as compared to its initial value? The electron gun–capacitor and capacitor–screen distances are both much greater than the length of the capacitor's plates.

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P132* A triangle is made from thin insulating rods of different lengths, and the rods are uniformly charged, i.e. the linear charge density on each rod is uniform and the same for all three rods. Find a particular point in the plane of the triangle at which the electric field strength is zero.

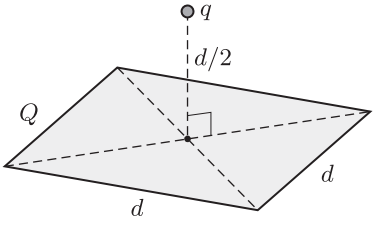
P133* Three very long (they can be regarded as infinitely long), thin, insulating rods are uniformly charged, i.e. the linear charge density on each rod is uniform and the same for all three rods. The rods lie in the same plane, and cross each other to make a general triangle. Where can a point charge be placed so as to be in equilibrium?

P134 Two very long, thin, insulating rods, each carrying uniform linear charge density λ, lie in perpendicular directions (in three-dimensional space), a distance d from each other (*see* figure). What is the magnitude of the force of repulsion between them?



P135 Two very long, parallel, thin, insulating rods are uniformly charged with equal but opposite electric charge densities. What shape are the electric field lines?

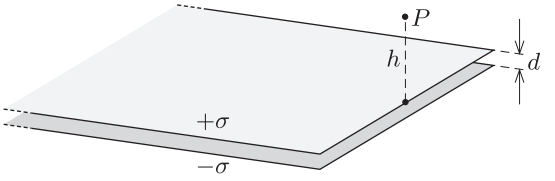
P136* A square of side d, made from a thin insulating plate, is uniformly charged and carries a total charge of Q. A point charge q is placed on the symmetrical normal axis of the square at a distance $d/2$ from the plate. How large is the force acting on the point charge?



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P137* A cube is made from six thin insulating square faces, each square having side d and carrying a uniformly distributed charge of Q. How large is the electrostatic force acting on each face?

P138** Two identical, large, rectangular, insulating plates lie one vertically above the other. The uniform separation between the thin plates is d, and they are both uniformly charged; the surface charge density on the upper one is $+\sigma$, and that on the lower one is $-\sigma$.

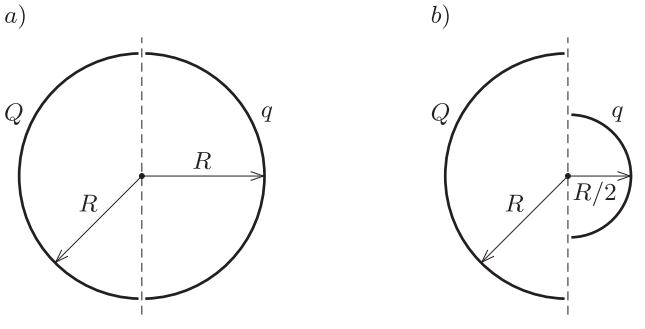


What is the magnitude and approximate direction of the electric field at the point P (*see* figure), which is at a height h vertically above the midpoint of one of the upper plate's edges?

The distance h is much smaller than the edge lengths of the insulating plates, but much larger than the separation d.

P139* There is a small circular hole in a thin-walled, uniformly charged, insulating spherical shell, which has radius R and carries a total charge Q. What is the electric field strength at the centre of the hole? Sketch the electric field lines associated with the holed spherical shell in a section that includes its axis of symmetry.

P140** Two insulating hemispherical shells (e.g. the two halves of a ping-pong ball) are placed very close to each other with their centres almost coincident, as shown in figure *a*). Both of them are uniformly charged, one with total electric charge Q, and the other with total charge q.



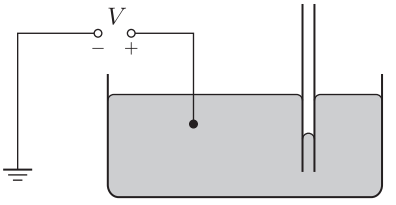
a) What is the magnitude of the net electric force exerted by the two bodies on each other?

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b) Does the result change if the radius of one of the hemispherical shells is only one-half that of the other, as shown in figure *b*)?

P141* Taking the zero of electrostatic potential at infinity, how many times larger is the potential at the centre of an insulating solid cube carrying uniform charge density than that at one of its vertices?

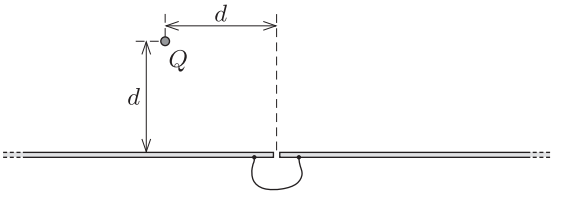
P142* A glass capillary tube is placed vertically into mercury held in a glass container (*see* figure). A high voltage, relative to ground, is applied to the mercury. Does the meniscus in the capillary tube (the interface between the mercury and air) move, and, if it does, which way?



P143 Near the edges of a parallel-plate capacitor, the electric field is inhomogeneous, but the effect of this so-called fringe field is usually neglected. Would a larger or smaller value be found for the magnitude of the capacitance if the fringing effect were also taken into account?

P144 One of the plates of a parallel-plate capacitor with capacitance C carries a charge of Q_1, and the other carries Q_2. What is the voltage (potential difference) across the capacitor?

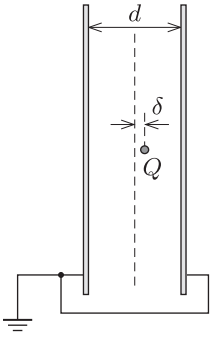
P145** Two very large, identical, rectangular, metal plates lie in the same plane, with corresponding edges parallel. The plates, which are very close to each other and connected by a wire, are initially uncharged. A point charge Q is now placed near them at the position above the plates shown in the figure. Find the resulting amounts of charge on each individual plate.



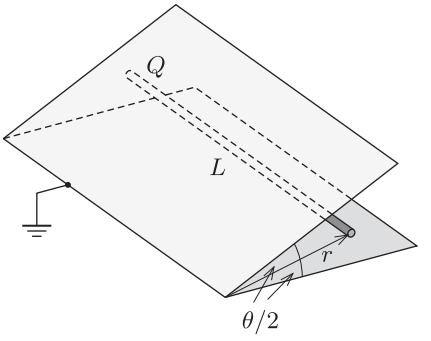
P146* Both plates of a parallel-plate capacitor, with plate separation d, are earthed, and a small pearl carrying charge Q is placed a distance δ ($\delta \ll d$) from

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the (imaginary) mid-plane between the plates (*see* figure). Find the force acting on the pearl.



P147* Two large earthed metal plates make an angle θ with each other. A thin rod of length L, and carrying a uniformly distributed charge Q, is placed in the plane bisecting the angle between the plates, parallel to, and at a distance r from, their line of the intersection ($L \gg r$).



Find the force that acts upon the rod as a result of the electrostatic induction *a*) if $\theta = 180°$ and *b*) if $\theta = 18°$.

P148** A thin metal disc of radius R is charged, and carries a total electric charge of Q. Find the charge distribution on the disc.

P149** Two thin metal discs, A and B, each of diameter 5 cm, are suspended by electrically insulating threads in such a way that the discs are parallel (*see* Fig. 1) and close to each other (for example, their separation might be 2 mm). Both discs have small charges q on them. As q is small, we may neglect the associated relative displacement of the discs, and the possibility of electrical discharge. The electrostatic force between the two discs is clearly repulsive at this stage.

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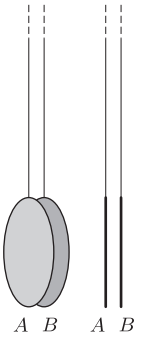


Fig. 1

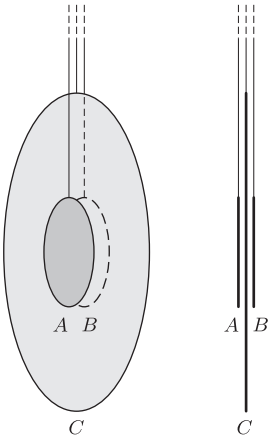
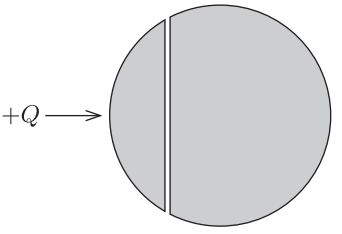


Fig. 2

A third metal disc C of diameter $D > 5$ cm is carefully placed between the two original discs. The third disc is uncharged and also suspended by an electrically insulating thread. The three discs are all parallel and their centres lie along the same horizontal line. The resulting set-up is shown in Fig. 2.

Find the diameter D of the third disc that will make the net electrostatic force acting on each charged disc equal to zero.

P150* A solid metal sphere of radius R is divided into two parts by a planar cut, made in such a way that the outer surface of the smaller part of the sphere is πR^2. The cut surfaces are coated with a negligibly thin insulating layer, and the two parts are put together again, so that the original shape of the sphere is restored. Initially the sphere is electrically neutral.



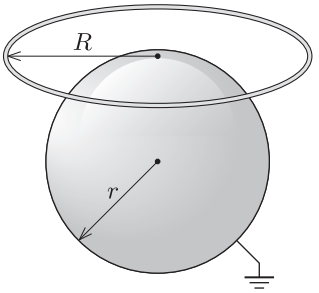
The smaller part of the sphere is now given a small positive electric charge $+Q$, while the larger part of the sphere remains neutral. Find

a) the charge distribution throughout the sphere,

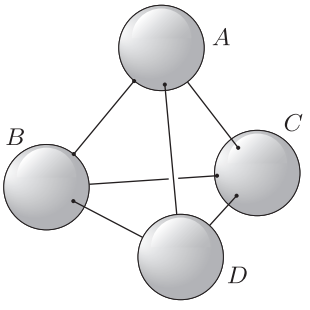
b) the electrostatic interaction force between the two pieces of the sphere.

P151 A thin metal ring of radius R is charged in such a way that the electric potential at its centre is V_0. The ring is now placed horizontally above a grounded solid metal sphere of radius r, so that the centre of the ring and the top of the sphere coincide. Find the total charge induced on the sphere.

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P152* Four identical non-touching metal spheres are positioned at the vertices of a regular tetrahedron, as shown in the figure. A charge of 20 nC given to one of the spheres, A, raises it to a potential V. Sphere A can also be raised to potential V if it and one of the other spheres are each charged with 15 nC.



What must be the size of equal charges given to A and two other spheres for the potential of A to again be raised to V? And if all four spheres are used?

P153* A small pearl carrying charge Q is placed at a distance d from the centre of a thin-walled spherical metal shell of radius R, with $d > R$. Find the force acting on the pearl if the shell is *a*) grounded, *b*) uncharged and not grounded, and *c*) charged, with charge Q'.

What would be the force acting on the pearl, in each of the three cases, if it were placed inside the spherical shell, at a distance $d < R$ from its centre?[11]

P154** An uncharged metal sphere of radius R is placed in a homogeneous electric field of strength E_0.

a) What is the electric field formed around the metal sphere?

b) Find the charge distribution on the sphere.

P155 Two small, identical bar magnets are placed some distance apart, as shown in the figure.

[11] The essence of this final complementary question can also be found in the predecessor of this book: see 'Problem 92' in P. Gnädig, G. Honyek & K. F. Riley, *200 Puzzling Physics Problems* (Cambridge University Press, 2001). Our reason for repeating its solution here is to present in one place all of the possible variants of this question.

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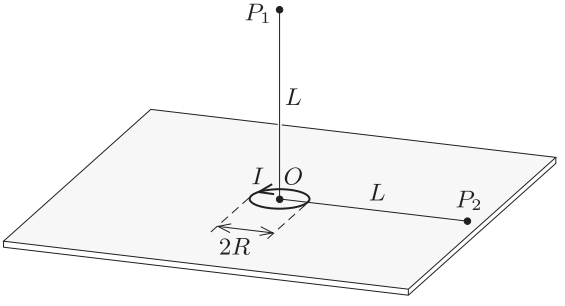


In which of the following cases is the work needed larger? And how many times larger?

a) The axis of the right-hand magnet is slowly rotated through 180°, so that the two magnetic moments are parallel.

b) The right-hand magnet is slowly removed 'to infinity' along the straight line connecting the magnets.

P156* A circular wire loop of radius R, and carrying an electric current I, is at rest in a horizontal plane (*see* figure).

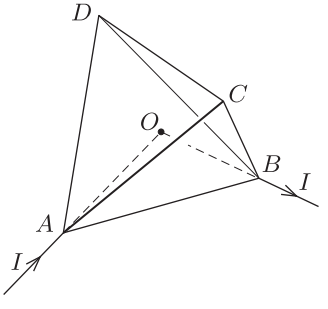


Find the magnetic field strength at a distance L ($\gg R$) from the centre O of the circular loop:

a) at a point, P_1, situated on the vertical axis passing through O,

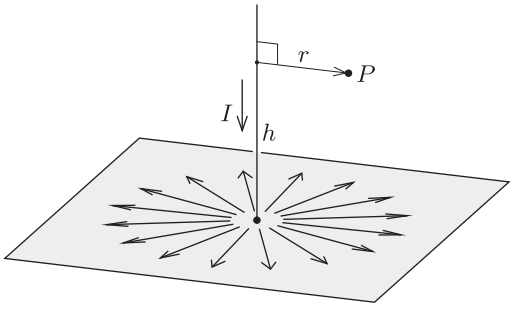
b) at a point, P_2 in the plane of the current loop.

P157* A regular tetrahedron is made of homogeneous resistance wire of uniform cross-section. Current I is conducted into vertex A through a long, straight wire directed towards the centre O of the tetrahedron, and it is conducted away through vertex B in the same way, as illustrated in the figure. What are the magnitude and direction of the magnetic field vector at the centre of the tetrahedron?



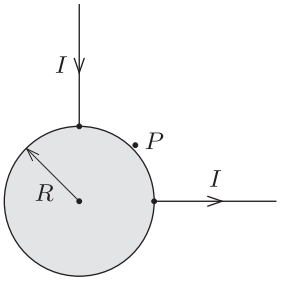
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P158* One end of a long, straight, current-carrying metal wire is electrically connected to a very large, thin, homogeneous metal plate, whose plane is perpendicular to the wire, and whose distant perimeter is earthed. One of the terminals of the battery supplying the current is also earthed.



Ignoring the Earth's field, find the magnitude and direction of the magnetic field at a point P that is a distance r from the wire and at a height h above the metal plate, when a steady current I flows in the wire (*see* figure). Describe the magnetic field above and below the plate.

P159** The surface of a plastic globe, of radius R, is uniformly covered with conducting material (e.g. with a graphite layer), and the globe is placed on an insulating support. One end of a long, straight, radial, current-carrying metal wire is electrically connected to a point on the sphere's surface. The steady current I, flowing through the surface, leaves the globe through another long, straight, radial metal wire that is perpendicular to the input wire, as shown.

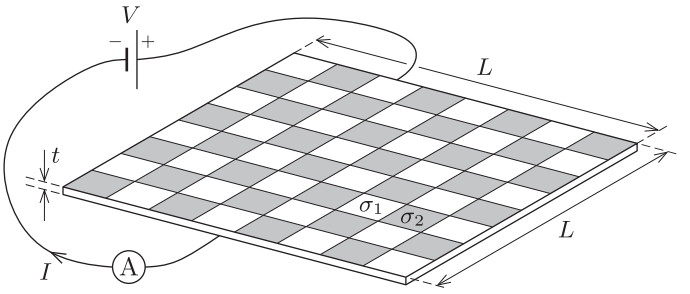


What kind of magnetic field is formed inside and outside the globe? Find, in particular, the magnetic field strength at the point P 'half-way' between the input and output junctions of the current, and just a 'whisker' above the globe's surface.

P160** Two very long, parallel, straight, thin wires, a distance d apart, carry electric currents with identical magnitudes but opposite directions. Find the shape of the associated magnetic field lines.

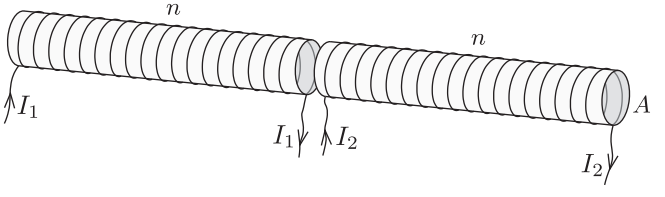
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P161*** A normal 8×8 chessboard is made from plates of two different metals, both of which are quite poor electrical conductors. The only other conducting elements in its construction are two thin terminal strips, which have very good conductivity. They are positioned one at each end of the board (but not shown in the figure). The common thickness t of the plates is much less than the length L of the board.



The conductivity of the light squares is σ_1, and that of the dark ones is σ_2. Find the current flowing through the chessboard, if a steady voltage V is applied across the terminals. Any interface resistances between the squares can be neglected.

P162* Two long, cylindrical, unifilar, air-cored solenoids are placed end-to-end and very close to each other, so that they have a common axis of symmetry (*see* figure). The solenoids are identical, with cross-sectional area A and n turns per unit length. The direct current flowing in one of them is I_1, and that in the other is I_2. What is the magnitude of the magnetic force between them?



P163* In a strong magnetic field B, the two ends of a thin, flexible wire of length ℓ are fixed at points P_1 and P_2, a distance $\ell/2$ apart. The direct current flowing in the wire is I. What is the shape of the wire, if the magnetic field vector is

a) perpendicular to the line segment P_1P_2,

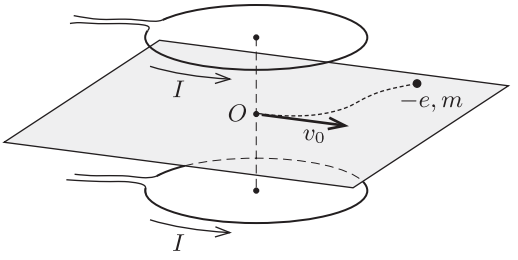
b) parallel to the line segment P_1P_2?

With what force does the wire pull on the anchor points in each case?

P164* Two identical circular wire loops are placed in parallel planes, with their centres on a common normal to both planes. The direct currents they carry are the same, both in magnitude I and in direction of flow (*see* figure). From the

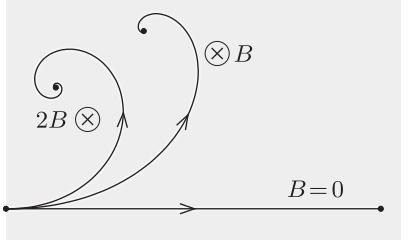
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midpoint O of the common normal, an electron with initial speed v_0 starts off in a direction parallel to the planes of the loops.



What can be said about the electron's speed and direction when it is very far from the loops?

P165* A charged particle enters a region in which there is a frictional force proportional to the particle's speed, and the particle stops 10 cm from its entry point. If the particle repeats its motion when a homogeneous magnetic field, perpendicular to the plane of its trajectory, is also present, then the particle comes to rest 6 cm from its entry point (*see* figure). How far from its entry point would the particle stop if the magnetic field were twice as large?



P166* A small ball of mass m, and carrying a positive charge q, is suspended by an insulating string of length ℓ. The pendulum so formed is placed, at rest, in a homogeneous, vertical magnetic field of strength B. Experiment shows that, if the ball is initially knocked slightly sideways, then it swings back and forth, with the plane of its swing slowly rotating. How long does it take for the plane to make one complete revolution?

P167** This problem investigates the motion of two electrons that are moving in a plane perpendicular to the field lines of a homogeneous magnetic field. The electrons are considered as classical point masses, affected only by electric and magnetic forces.

a) The two electrons, initially at rest, are placed a distance d apart. They are then given initial velocities of identical magnitudes v, but in opposite directions. Find the condition that d must satisfy if, in the subsequent motion, the separation is to remain constant. Find also an expression for v.

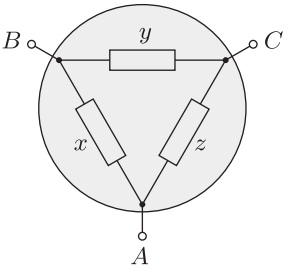
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b) Show that it is possible to maintain a constant separation d if only one of the two electrons is given an initial velocity. What is the trajectory of the centre of mass of the system in this case? Find the minimal distance $d_{\min}$ that is necessary to realise this kind of motion. Sketch the trajectories of the electrons in that case. When will the initially moving electron first stop?

P168 The electrical resistances measured across the three pairs of terminals of the 'transparent' black box shown in the figure are

$$AB\text{: } 1\ \Omega, \qquad BC\text{: } 2\ \Omega, \qquad AC\text{: } 3\ \Omega.$$

Find the values of resistors x, y and z.



We can write the following simultaneous equations expressing the unknown resistances (in units of ohms) by means of their labels:

$$\frac{1}{x}+\frac{1}{y+z}=1 \quad \longrightarrow \quad x(y+z)=x+y+z, \tag{1}$$

$$\frac{1}{y}+\frac{1}{x+z}=\frac{1}{2} \quad \longrightarrow \quad y(x+z)=2(x+y+z), \tag{2}$$

$$\frac{1}{z}+\frac{1}{x+y}=\frac{1}{3} \quad \longrightarrow \quad z(x+y)=3(x+y+z). \tag{3}$$

Subtracting equation (3) from the sum of (1) and (2), we get $2xy = 0$. This is *impossible*, because neither x nor y is zero!

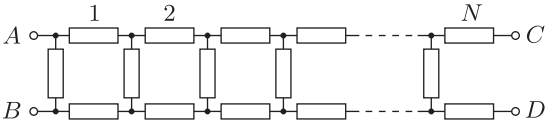
How can this mathematical 'contradiction' be resolved?

P169* Closed strips are made from the chain shown in the figure, which consists of $3N$ identical resistors, in two different ways:

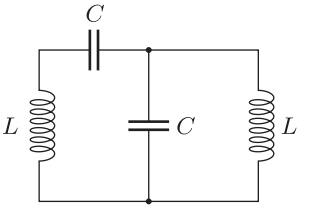
a) the terminals A and C, and the terminals B and D, are connected pairwise (ordinary strip),
b) the terminals A and D, and the terminals B and C, are connected pairwise (Möbius strip).

In which of the two cases is the equivalent resistance between points A and B the larger?

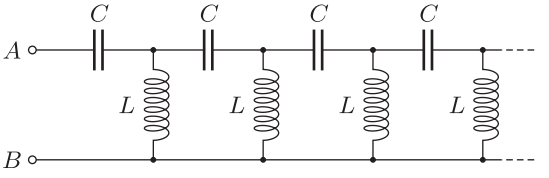
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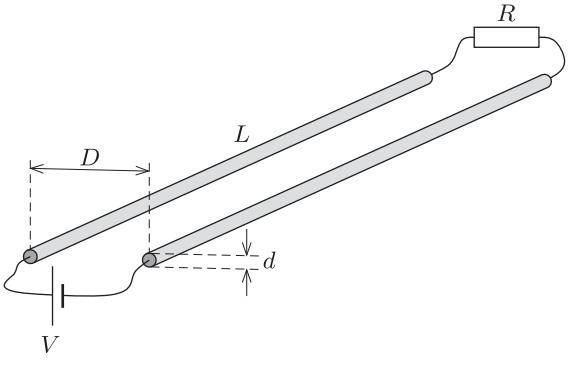
P170 Find the natural frequencies of an LC circuit with two identical inductors and two identical capacitors (a '2L2C' circuit), connected as shown in the figure.



P171** Find the equivalent electrical impedance between the terminals A and B shown in the figure, for an alternating current of frequency ω. The 'infinite' chain is made up of a large number of identical units, each consisting of a coil of inductance L and a capacitor with capacitance C. Is it possible that the equivalent impedance has two different values?

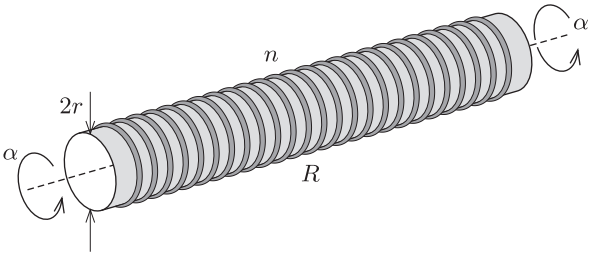


P172* Two identical, very long, cylindrical conductors, of diameter d and negligible resistance, are placed parallel to each other with their axes separated by $D = 50d$. A battery of electromotive force (voltage) V is connected between the left-hand ends of the wires, while a resistor with resistance R is connected across their other ends (*see* figure). Find the resistance R that makes the electrical and magnetic forces between the conductors equal.



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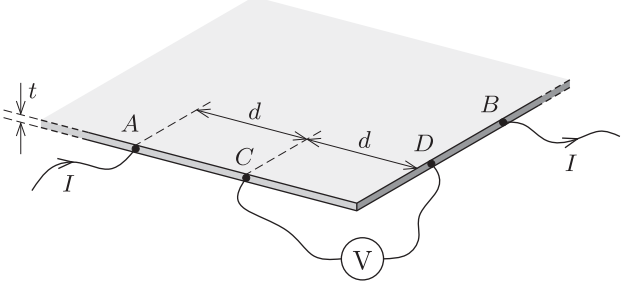
P173* In 1917, T. D. Stewart and R. C. Tolman discovered that an electric current flows in any coil wound around, and attached to, a cylinder that is rotated axially with constant angular acceleration.



Consider a large number of rings of thin metallic wire, each with radius r and resistance R. The rings have been glued in a uniform way onto a very long evacuated glass cylinder, with n rings per unit length of the symmetry axis. The plane of each ring is perpendicular to that axis.

At some particular moment, the cylinder starts to accelerate around its symmetry axis with angular acceleration α. After a certain length of time, there is a constant magnetic field B at the centre of the cylinder. Find, in terms of the charge e and mass m of an electron, the magnitude of the field.

P174* We aim to measure the resistivity of the material of a large, thin, homogeneous square metal plate, of which only one corner is accessible. To do this, we chose points A, B, C and D on the side edges of the plate that form the corner (*see* figure). Points A and B are both $2d$ from the corner, whereas C and D are each a distance d from it. The length of the plate's sides is much greater than d, which, in turn, is much greater than the thickness t of the plate.



If a current I enters the plate at point A, and leaves it at B, then the reading on a voltmeter connected between C and D is V. Find an expression for the resistivity ϱ of the plate material.

P175 A metal sphere is electrically charged and hangs on an insulating cord. The sphere slowly loses its charge because the air has a small, but non-zero, con-

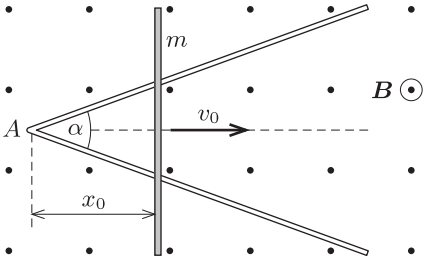
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ductivity σ. Assuming that the air's conductivity is everywhere the same, how long will it take for the charge on the sphere to halve?

P176* A chocolate figure of Santa Claus, wrapped in aluminium foil, is electrically charged and hangs on an insulating cord. The figure slowly loses its charge because the air has a small, but non-zero, conductivity σ. Assuming that the air's conductivity is everywhere the same, how long will it take for Santa's charge to halve?



P177 A long, straight wire of negligible resistance is bent into a V shape, its two arms making an angle α with each other, and placed horizontally in a vertical, homogeneous magnetic field of strength B. A rod of total mass m, and resistance r per unit length, is placed on the V-shaped conductor, at a distance x_0 from its vertex A, and perpendicular to the bisector of the angle α (*see* figure).



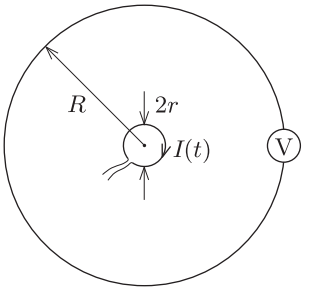
The rod is started off with an initial velocity v_0 in the direction of the bisector, and away from A. The rod is long enough not to fall off the wire during the subsequent motion, and the electrical contact between the two is good – although the friction between them is negligible.

Where does the rod ultimately stop?

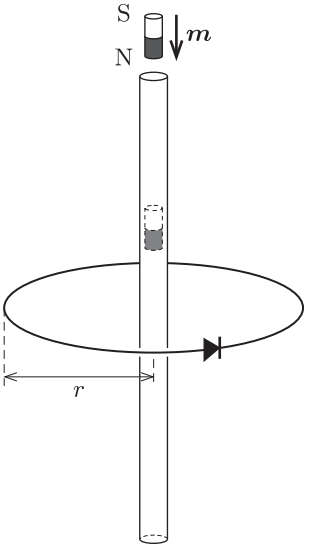
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P178 Inside a vertical, thin-walled, non-ferromagnetic (say, brass) tube, a quite large, strong, cylindrical permanent (say, neodymium) magnet falls very slowly. It takes a time t_1 to fall between two particular markers. If the experiment is repeated with a different non-ferromagnetic (say, copper) tube with the same length but a slightly larger diameter, then the corresponding time is t_2. How long does it take for the magnet to fall between the marks when the two tubes are fitted inside each other? The mutual inductance between the tubes is negligible.

P179 Two circular wire loops, with radii R and r ($r \ll R$), are concentric and lie in the same plane (*see* figure). The electric current in the smaller loop is increased uniformly from zero to a value of I_0 over a time interval t_0. Find the induced voltage in the larger loop.

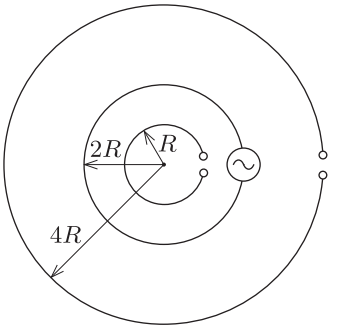


P180* A closed circular loop of radius r is made from a wire of resistance R and a diode, which can be considered ideal. The loop is held in a horizontal plane, and a long, vertical glass tube passes through its centre (*see* figure). Find the charge that flows through the diode if a small bar magnet with magnetic moment $\boldsymbol{m}$ falls through the tube.



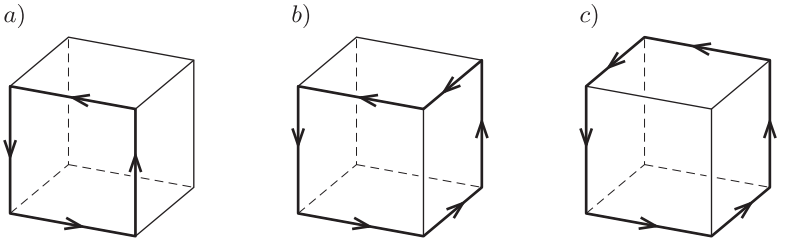
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P181* Three nearly complete circular loops, with radii R, $2R$ and $4R$, and made of thin wire, are placed concentrically on a horizontal table-top, as shown in the figure. A time-varying electric current is made to flow in the middle loop. Find the voltage induced in the largest loop at the moment when the voltage between the terminals of the smallest loop is V_0.



P182 Two identical superconducting rings are very far from each other. The current in one of the rings (say, A) is I_0, but there is no current in the other (B). The two rings are now slowly brought closer together. Find the current that flows in A when that in B has increased to I_1.

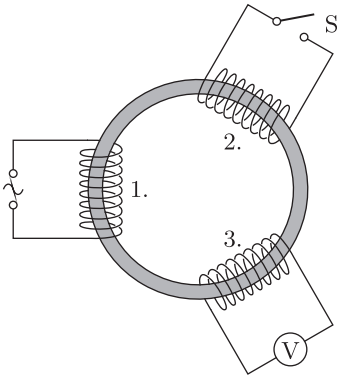
P183* Three metal wires of lengths $4a$, $6a$ and $6a$ are arranged along the edges of a cube of side a in three different ways, as shown in the figure. The coefficient of self-inductance of the square shown in figure a) is measured as L_1, and for the arrangement shown in figure b) it is L_2. How large, expressed in terms of L_1 and L_2, would it be for the arrangement shown in figure c)?



P184* Three identical, 'wide' electromagnetic coils, of negligible ohmic resistance, are wound onto a 'narrow' toroidal (doughnut-shaped) iron core, equally spaced around its circumference, as shown in the figure. The first coil is connected to an (ideal) alternating-current power supply, the second to an open switch S, and the third to a very high-resistance voltmeter. With this

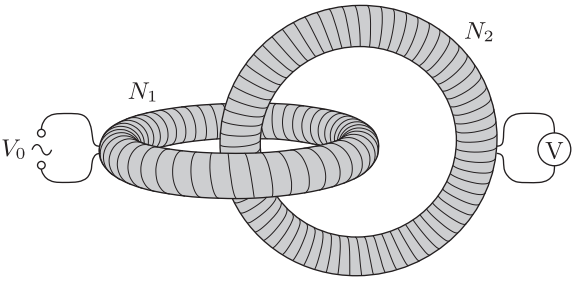
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arrangement, the voltmeter shows a root mean square (r.m.s.) reading equal to one-half of that of the power supply.



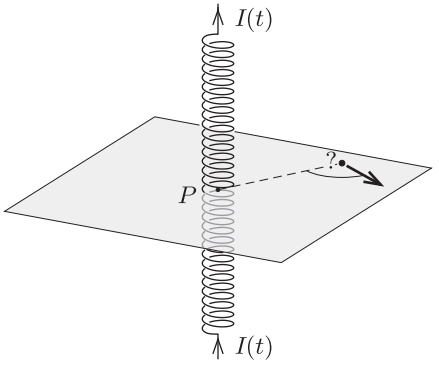
At this point, switch S is closed and short-circuits the terminals of the second coil. Assuming that the magnetic permeability of the iron core does not depend on the magnetic flux through it, determine the new reading of the voltmeter.

P185** Two ideal (with zero ohmic resistance) air-core toroidal coils, of identical size but with different numbers of turns, N_1 and N_2, are interlinked, as shown in the figure. The planes containing their centres and major radii are perpendicular to each other. The terminals of the coil with N_1 turns are connected to the normal household a.c. supply with r.m.s. voltage V_0, and an ideal voltmeter is connected to those of the other coil. Find the reading on the voltmeter.



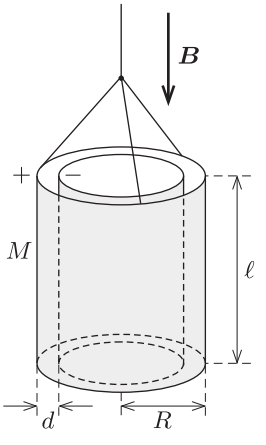
P186* A small, electrically charged pearl, initially at rest, can move on a horizontal, frictionless plane. Not far from the pearl, there is a long, vertical solenoid, in which the electric current is first increased uniformly from zero to a given value, and then uniformly decreased back to zero. In which direction, relative to point P in the figure, will the pearl have moved by the end of the process? Or will it (still) be where it started?

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P187* The current in an air-core, toroidal coil with major radius 0.1 m, cross-sectional area 2 cm^2 and 200 turns changes uniformly (over a short interval) at a rate of 10 A s^{-1}. Find the initial acceleration of a proton, treated as a classical point mass, that starts from rest at the centre of the toroid.

P188* A cylindrical capacitor with external radius R, internal radius $R - d$ ($d \ll R$), length ℓ and mass M hangs on an insulating cord in a region where there is a homogeneous, vertical magnetic field of strength B. It can rotate freely (as a whole) around its vertical axis, but is constrained so that it cannot move horizontally. The capacitor is charged and there is a voltage difference V between its plates.



What happens, if:

a) without being mechanically disturbed, the capacitor is discharged through an internal radial wire,

b) suddenly, the magnetic field is switched off?

P189* It is well known that, in a homogeneous magnetic field $\boldsymbol{B}_0$, an observer moving with velocity $\boldsymbol{v}_0$ ($|\boldsymbol{v}_0| \ll c$) 'experiences' an electric field, given by $\boldsymbol{E}_0 = \boldsymbol{v}_0 \times \boldsymbol{B}_0$. This phenomenon is called motional electromagnetic induction.

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Using only physical laws known to high-school students, investigate whether or not the inverse phenomenon exists. Does an observer 'experience' a magnetic field when moving in a homogeneous electric field?

P190** A parallel-plate capacitor, with its plates vertical, is charged in the sense shown in Fig. 1, and positioned with its lower edges above and on either side of a small horizontal compass needle. The capacitor is discharged when the tops of the plates are joined using a small conducting rod. Describe the response of the compass needle during the discharge process.

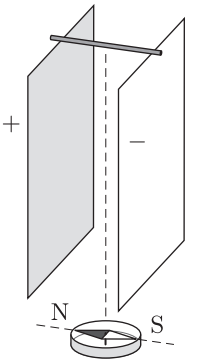


Fig. 1

This problem was used as a question in a Hungarian national physics competition, several years ago. The 'official solution' to the problem read more or less as follows:

> When the tops of the plates are joined by the conducting rod, an electric current starts to flow in the rod, from left to right, with magnitude $I = -\mathrm{d}Q/\mathrm{d}t$. At the same time, the electric field between the plates $E = \varepsilon_0^{-1} Q/A$ changes, and J. C. Maxwell has shown that this generates the so-called displacement current. The size of the current is proportional to the rate of change of the electric field, with proportionality constant ε_0.
>
> The displacement current density between the plates is
>
> $$j_D = \varepsilon_0 \dot{E} = \varepsilon_0 \frac{1}{\varepsilon_0} \frac{1}{A} \frac{\mathrm{d}Q}{\mathrm{d}t} = -\frac{I}{A}.$$
>
> The current in the rod (of magnitude I) and the displacement current (of magnitude $j_D A = -I$), taken together, can be considered as charge moving in a closed circuit.
>
> According to Maxwell's equations, magnetic fields are produced not only by ordinary electric currents, but also by displacement currents. As the displacement current and the current in the rod have equal magnitudes (but opposite directions),

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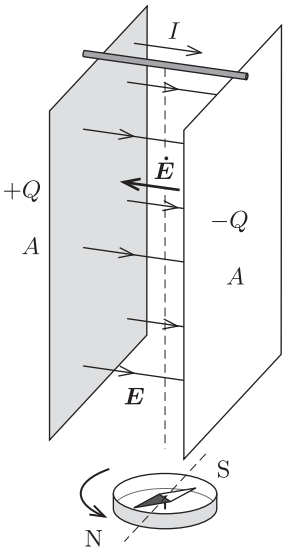


Fig. 2

and the displacement current is closer to the compass needle, its effect is larger. This is why, during the discharge process, the compass needle tends to rotate (*see* Fig. 2), with its north pole moving in an anticlockwise direction (from the position shown) when viewed from above. It will subsequently return to its original position.

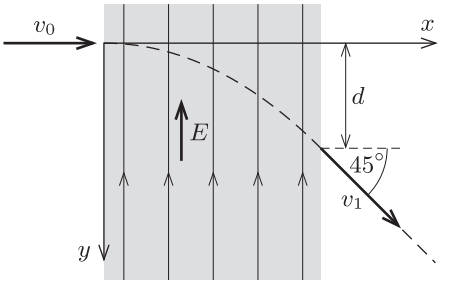
Decide, whether the 'official solution' to the problem is right or wrong.

P191* A chocolate figure of Santa Claus, wrapped in aluminium foil, is electrically charged and hangs on an insulating cord. The figure slowly loses its charge because the air has a small, but non-zero, conductivity. Describe the magnetic field around Santa during the discharge process.



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P192 An electron, moving with 60 % of the speed of light c, enters a homogeneous electric field that is perpendicular to its velocity. When the electron leaves the field, its velocity makes an angle of 45° with its initial direction (*see* figure).



a) Find the speed v_1 of the electron after it has crossed the electric field.

b) Find the distance d shown in the figure, if the strength of the electric field is $E = 510$ kV m^{-1}.

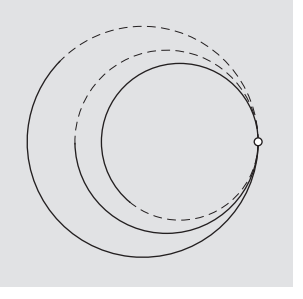
The rest energy, mc^2, of an electron is 510 keV.

P193 Imagine a circular evacuated tube running around the Earth's magnetic equator, in which – in principle – electrons and protons could orbit under the influence of the Earth's magnetic field.

a) Estimate the required speed for each particle, and determine the corresponding direction of circulation.

b) Express the particle energies needed for these hypothetical flights in eV units.

P194* The trajectories of charged particles, moving in a homogeneous magnetic field, can be followed by observing the trails they produce in cloud chambers (e.g. a Wilson chamber). Is it possible that, when a charged particle decays into two other charged particles, the trail segments close to the decay point (before the particles have started to slow down significantly) are arcs of circles that touch each other (as shown in the figure)?



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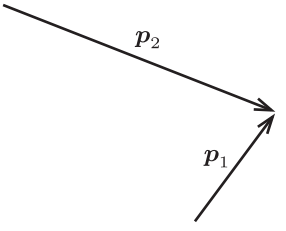
P195* The pion (π^+) is a subatomic particle with a mass 273 times larger than that of an electron. In one of its possible decay modes, it decays into a positron (e^+) and an electron-neutrino (ν_e):

$$\pi^+ \longrightarrow e^+ + \nu_e.$$

What is the minimum speed of the pion if, following its decay, the positron and the neutrino move at right angles to each other?

Assume that the neutrino has zero rest mass. For such a particle, the connection between its energy and linear momentum is $E = pc$.

P196* Investigate the elastic collision of two (ultra-relativistic) particles, moving with speeds very close to the speed of light. The linear-momentum vectors of the particles, before the collision, are shown in the figure. Determine the minimal possible angle between the directions of the particles after the collision.



P197 An electron orbits uniformly around a circular trajectory in a homogeneous magnetic field. Is it possible for the magnitude of the magnetic field produced by the electron's motion to be larger, at the centre of the circle, than that of the field?

P198* Estimate the pressure exerted on the walls of a small cubical box, which has edges of length d, by a neutron enclosed within the box.

P199 *Positronium* is a system consisting of an electron and its antiparticle, a positron, bound together into an exotic 'atom' by their electrostatic attraction.[12] The orbit and energy levels of the two particles are similar to those of the hydrogen atom. Find – within the framework of a Bohr model – the energies of positronium's ground and excited states.

P200 The fuel consumption of a small car is 4 litre/100 km. Convert the consumption into SI units and find an (imaginative) physical interpretation for the fuel consumption expressed in these units.

[12] A positronium atom is unstable; the two particles annihilate each other, producing predominantly two or three gamma photons.

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